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Kumar et al.

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(54) **MEASUREMENT REPORTING
ENHANCEMENTS IN BATCH MODE
REPORTING**

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H04L 5/00 (2006.01)
H04W 64/00 (2009.01)

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H04L 5/0048; H04L 5/0091; H04L
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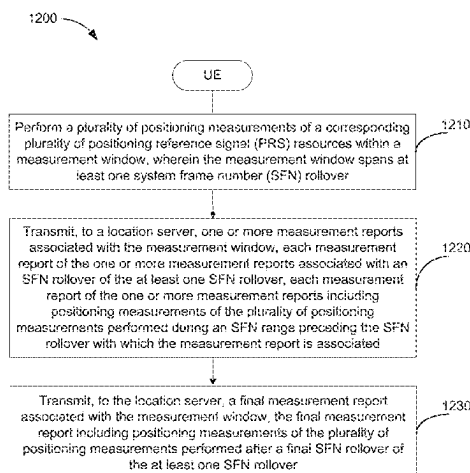
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(57) **ABSTRACT**

Disclosed are techniques for wireless positioning. In an
aspect, a user equipment (UE) performs a plurality of
positioning measurements of a corresponding plurality of
positioning reference signal (PRS) resources within a mea-
surement window, wherein the measurement window spans
at least one system frame number (SFN) rollover, and
transmits, to a location server, at least one measurement
report associated with the measurement window, the at least
one measurement report including a plurality of timestamp
information elements corresponding to the plurality of posi-
tioning measurements, wherein at least one timestamp infor-

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mation element of the plurality of timestamp information elements indicates the at least one SFN rollover.

15 Claims, 15 Drawing Sheets

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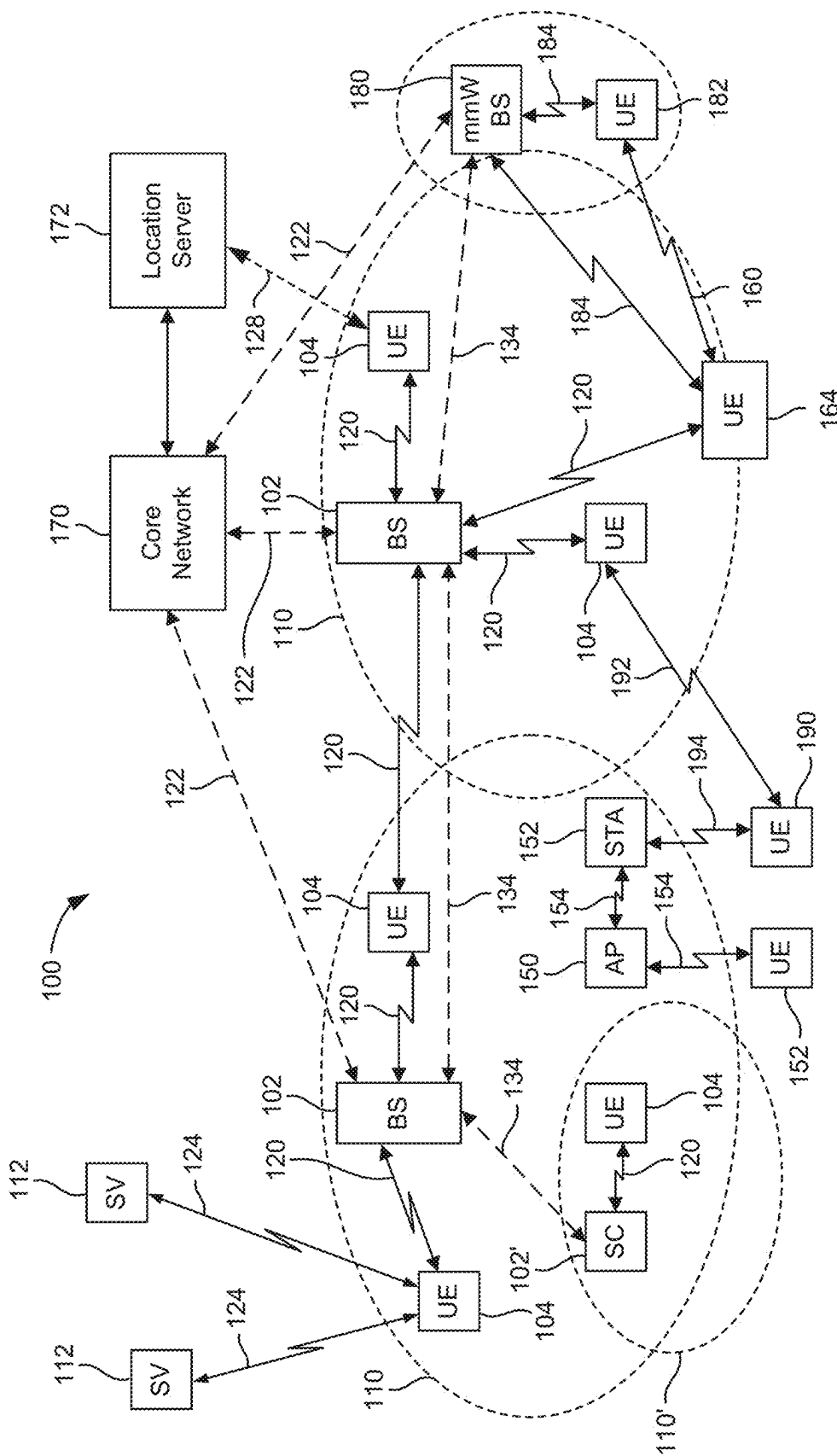


FIG. 1

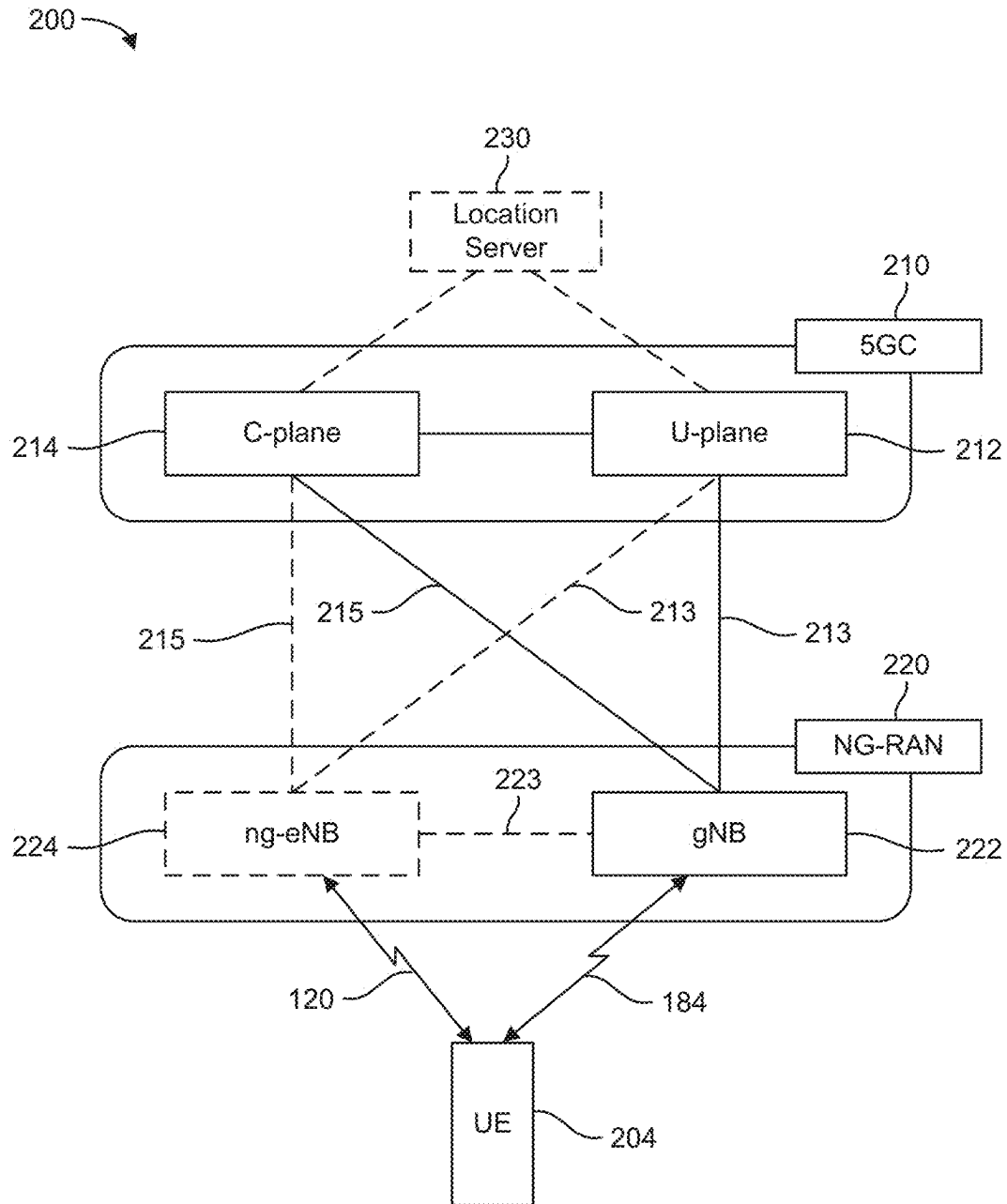


FIG. 2A

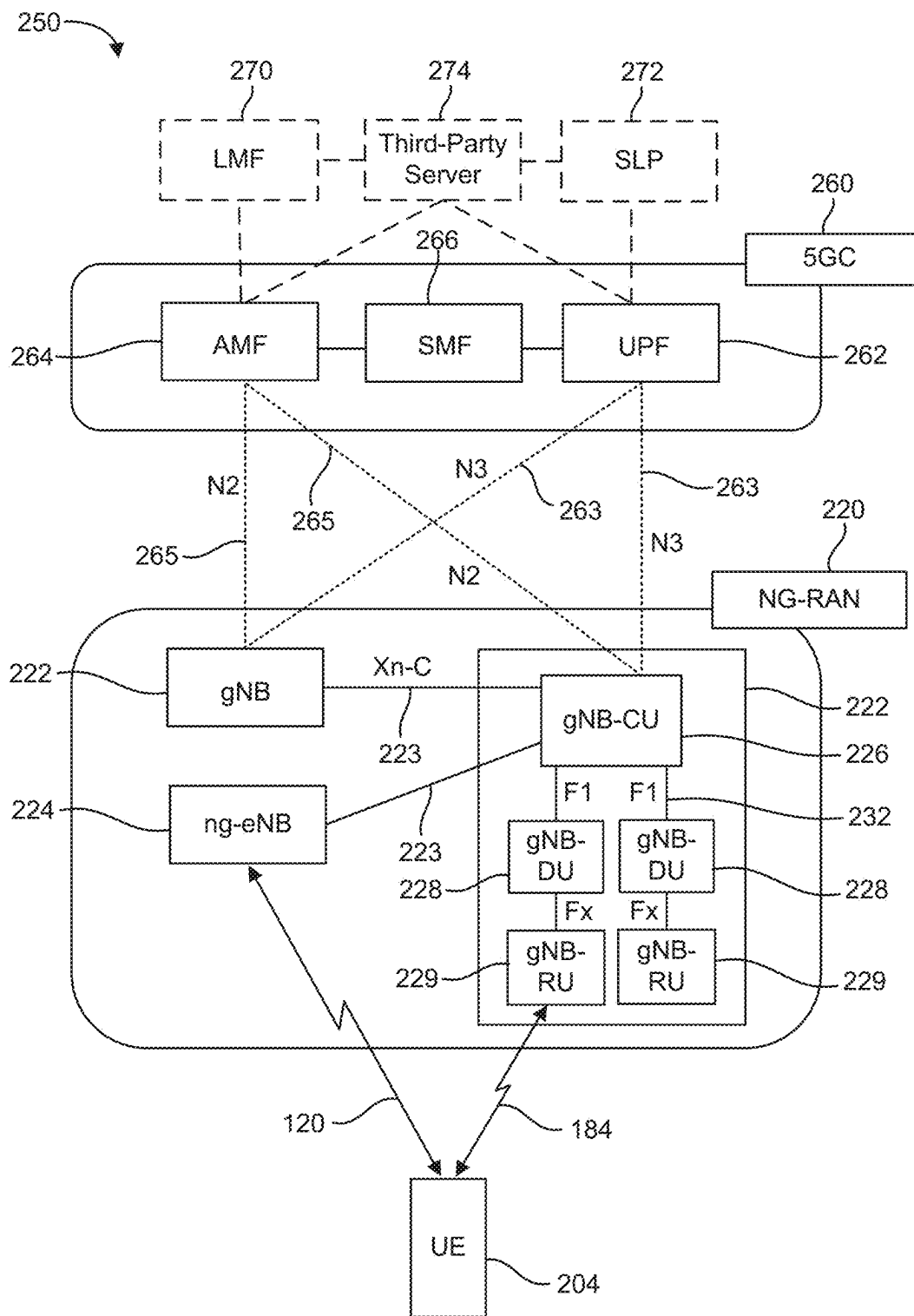


FIG. 2B

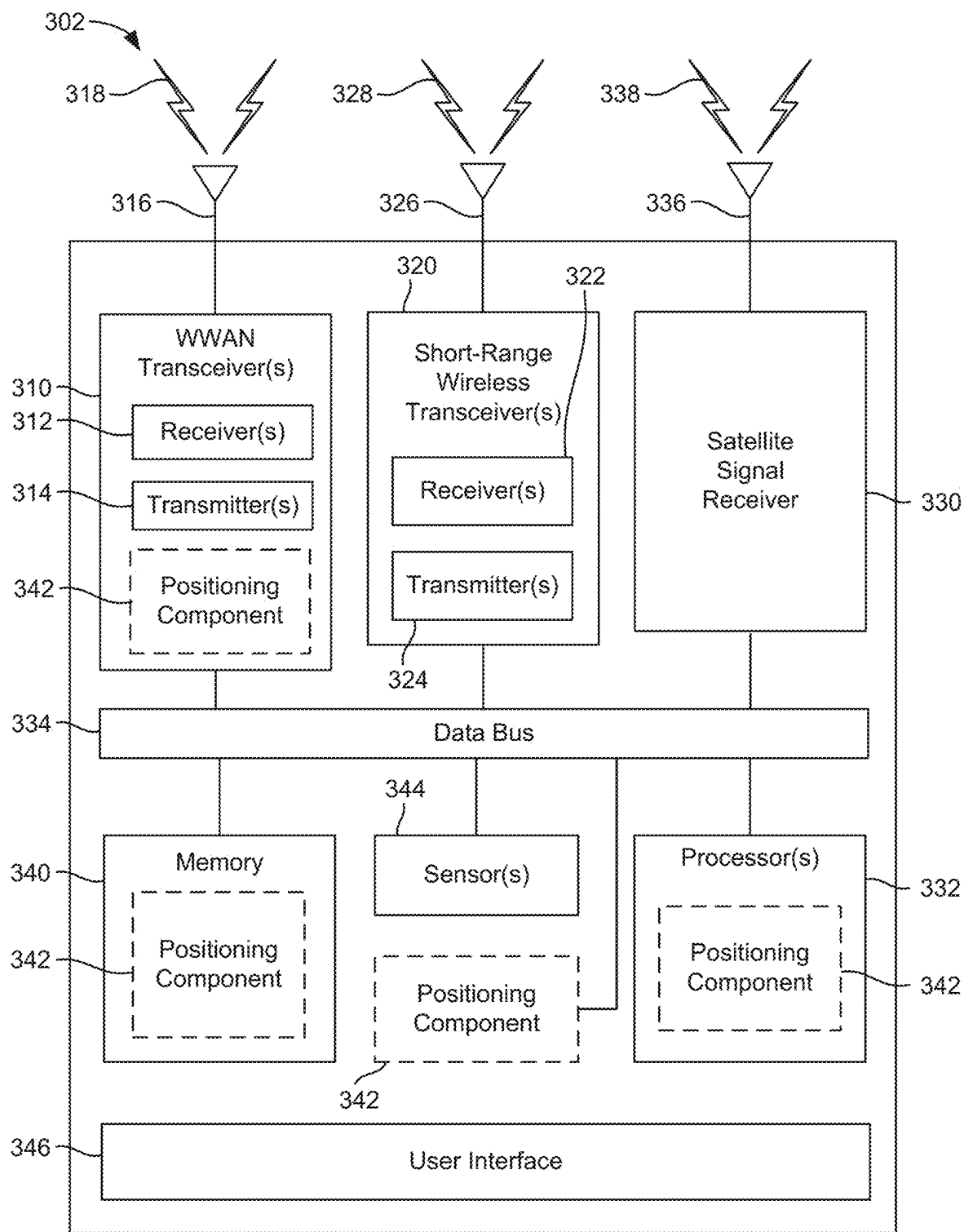


FIG. 3A

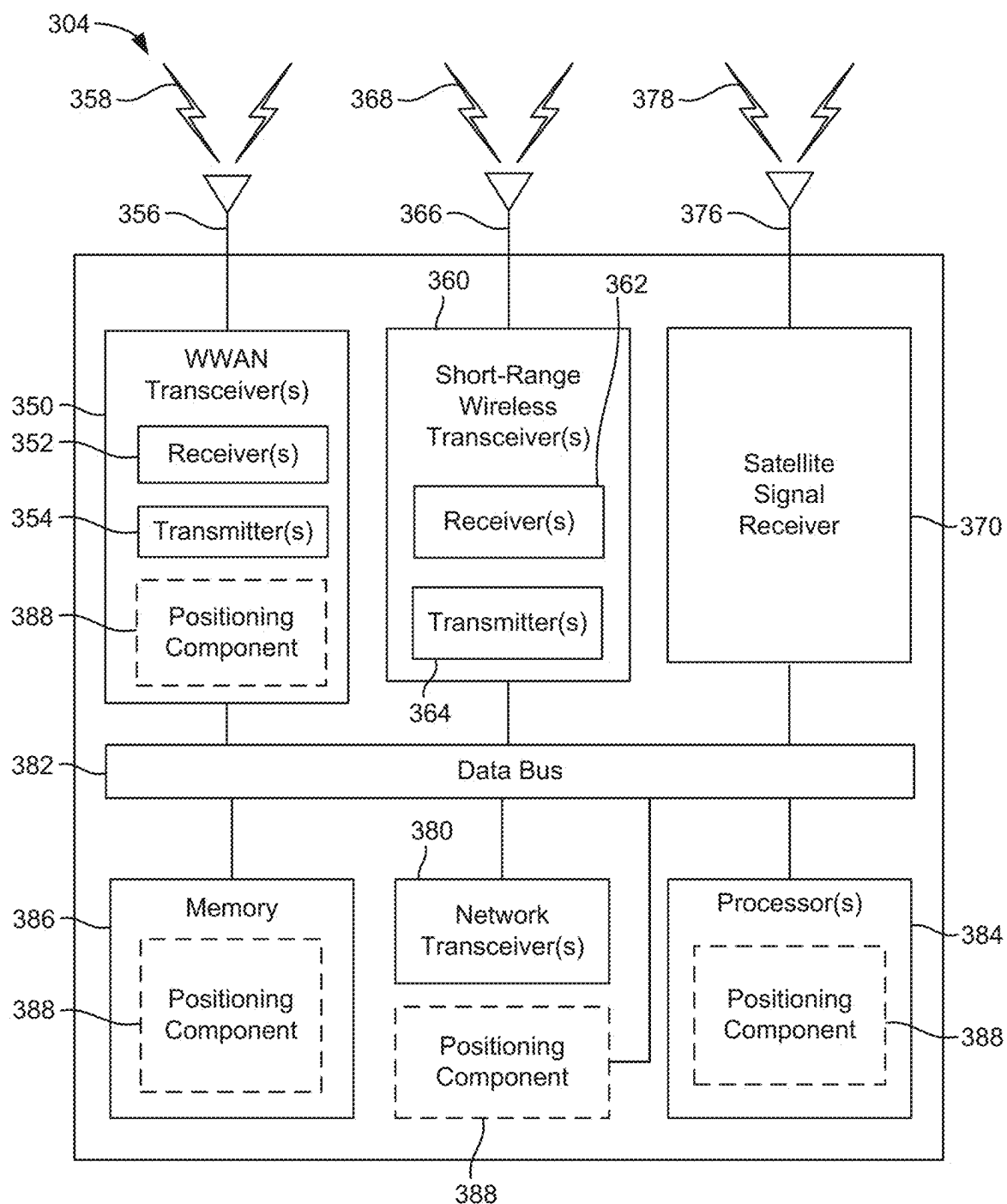
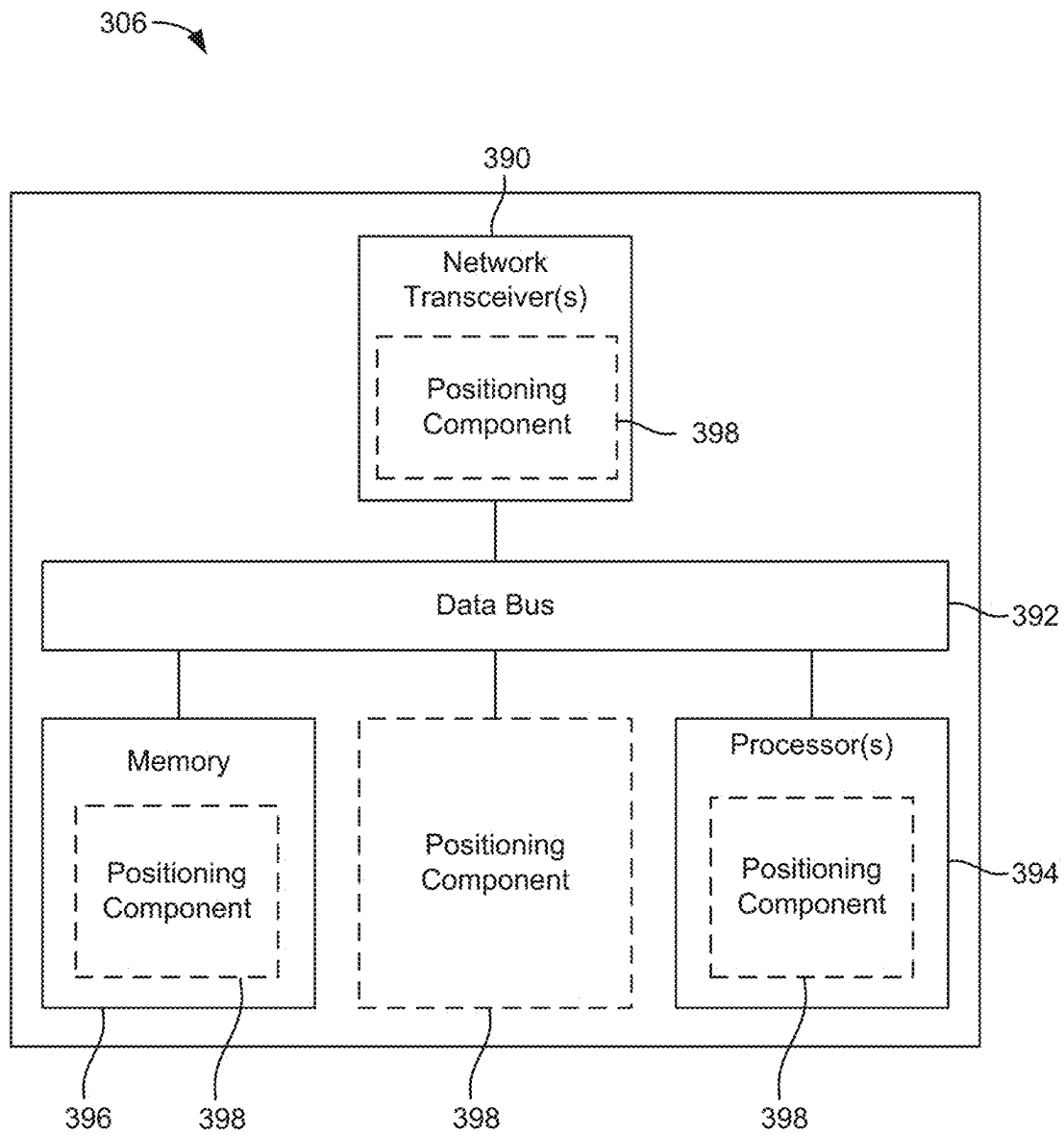


FIG. 3B

*FIG. 3C*

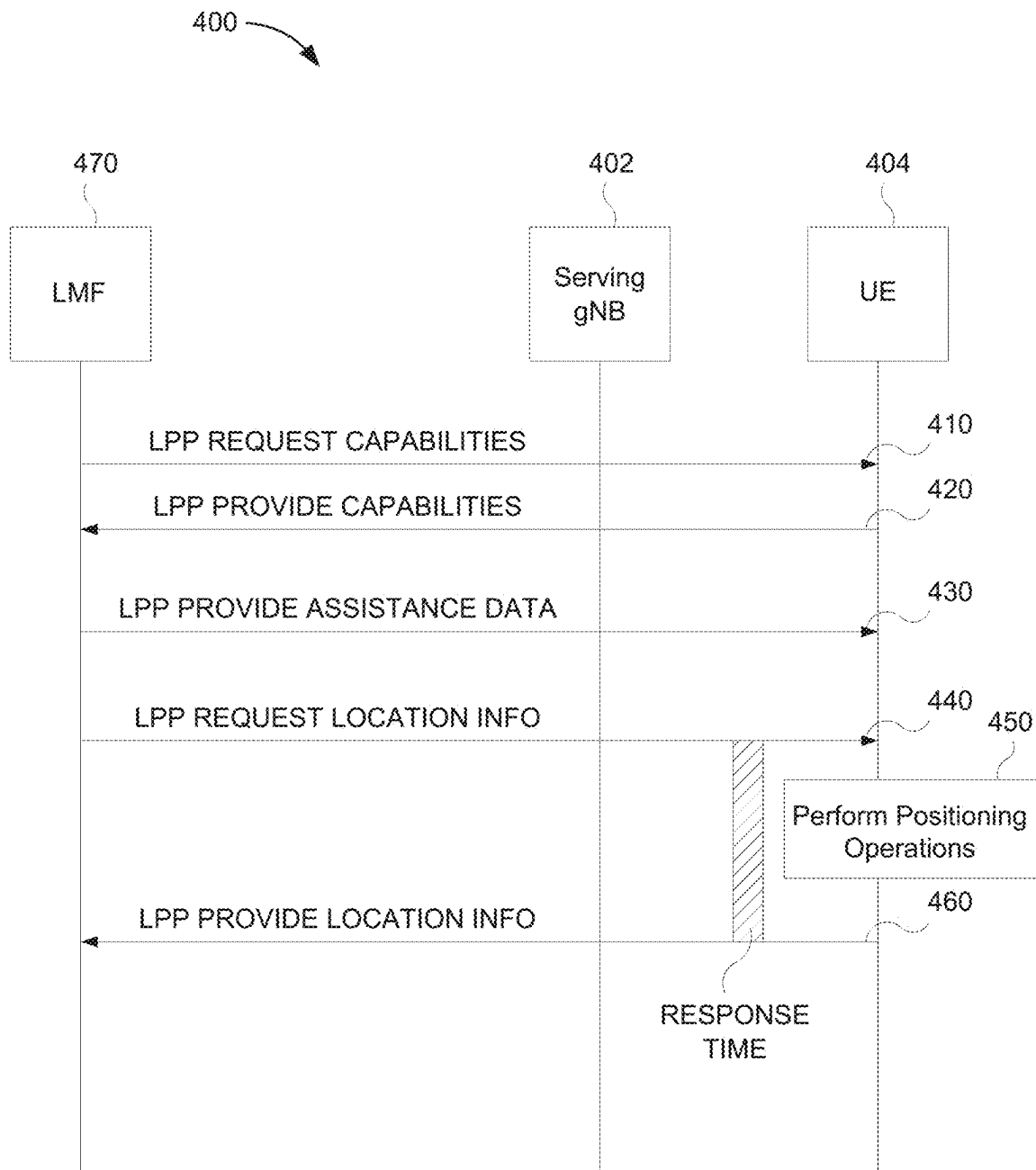


FIG. 4

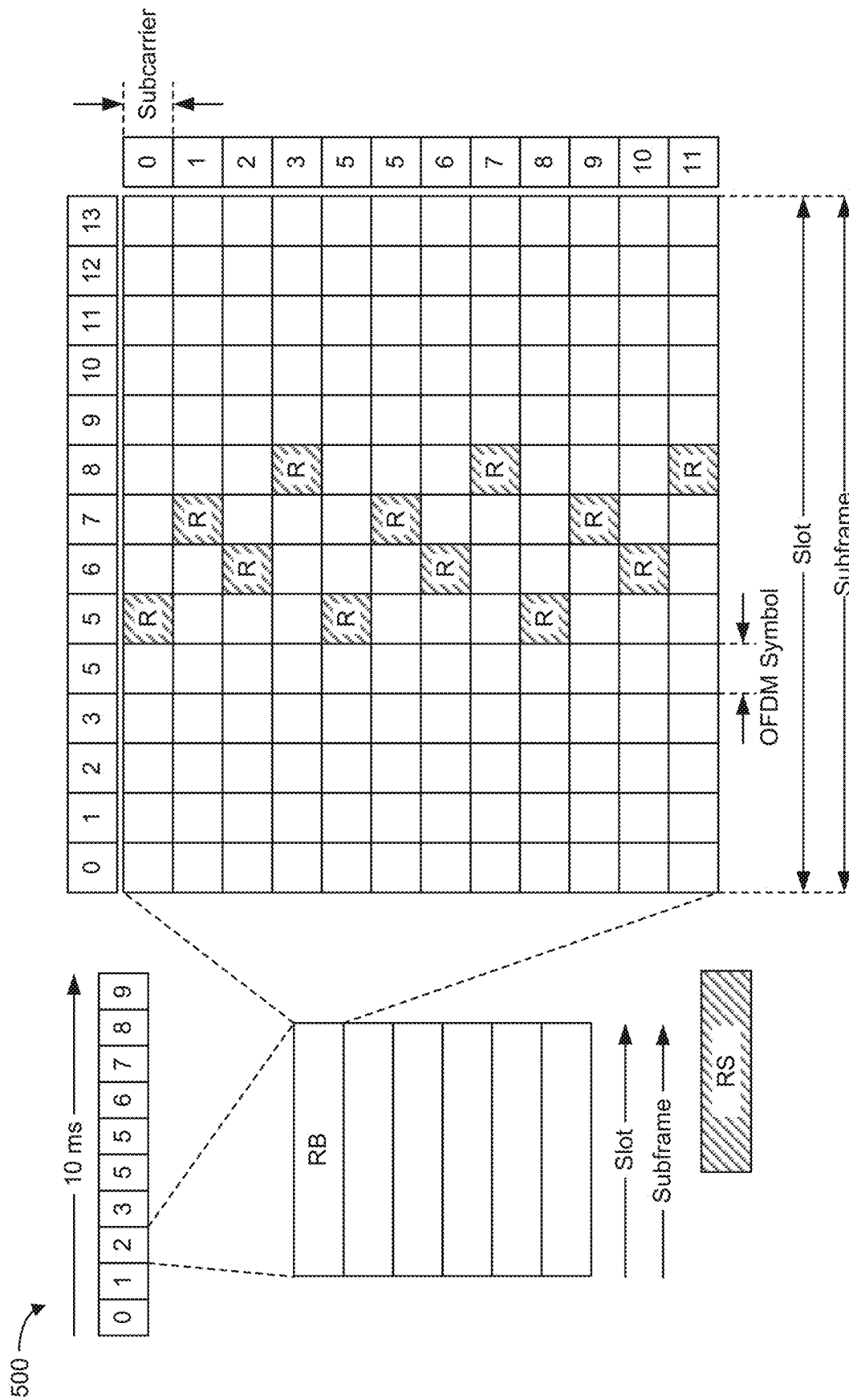


FIG. 5

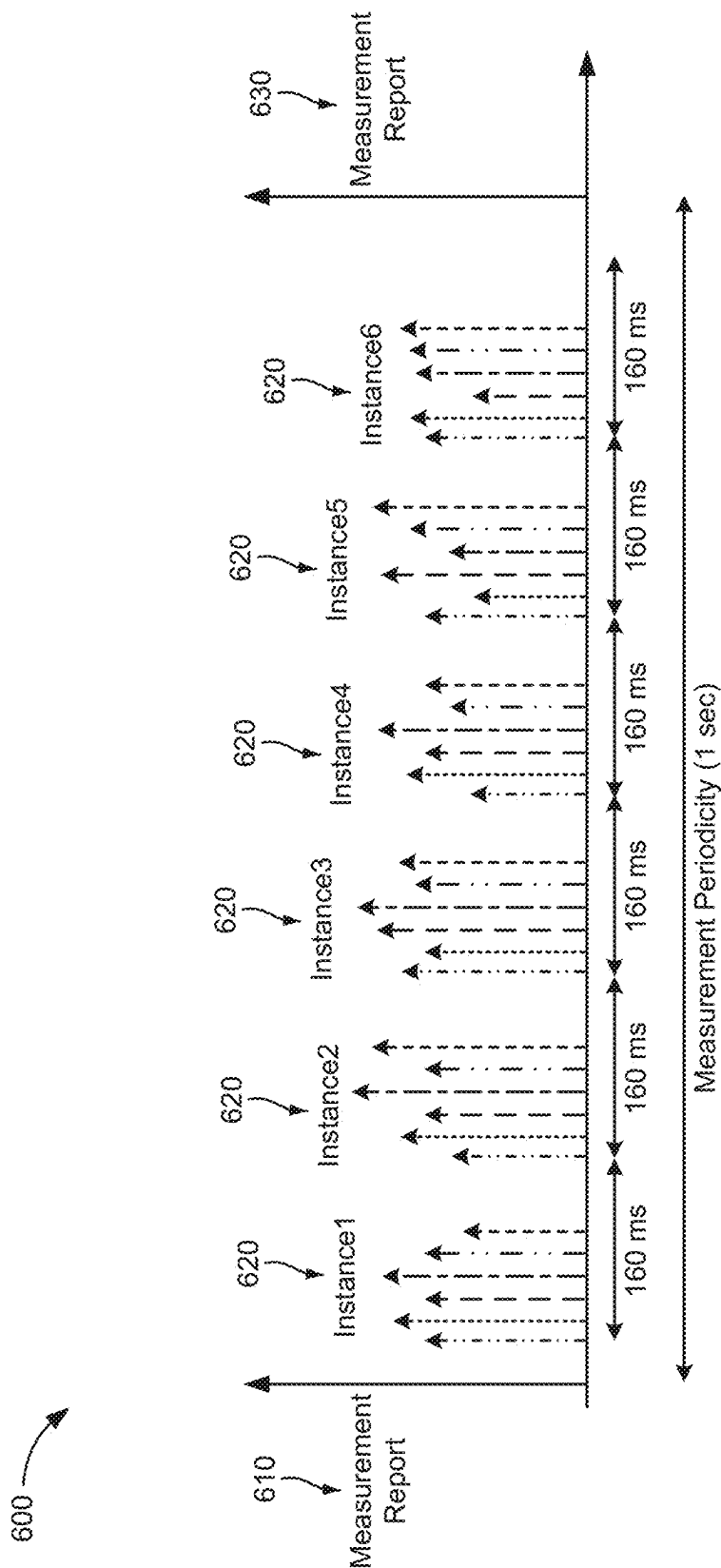


FIG. 6

700 

```
NR-TimeStamp-r16 ::= SEQUENCE {
    dl-PRS-ID-r16          INTEGER (0..255),
    nr-PhysCellID-r16      NR-PhysCellID-r16
    nr-CellGlobalID-r16    NCGI-r15
    nr-ARFCN-r16           ARFCN-ValueNR-r15
    [ nr-SFN-r16           INTEGER (0..1023), ]
    nr-Slot-r16            CHOICE {
        scs15-r16          INTEGER (0..9),
        scs30-r16          INTEGER (0..19),
        scs60-r16          INTEGER (0..39),
        scs120-r16         INTEGER (0..79)
    },
    ...
}
```

OPTIONAL,
OPTIONAL,
OPTIONAL,

FIG. 7

800

```

QoS ::= SEQUENCE {
    horizontalAccuracy          HorizontalAccuracy          OPTIONAL,
    verticalCoordinateRequest   BOOLEAN,
    verticalAccuracy            VerticalAccuracy            OPTIONAL,
    responseTime                ResponseTime                OPTIONAL,
    velocityRequest             BOOLEAN,
    ...,
    [[ responseTimeNB-r14       ResponseTimeNB-r14         OPTIONAL
    ]],
    [[ horizontalAccuracyExt-r15 HorizontalAccuracyExt-r15  OPTIONAL,
    verticalAccuracyExt-r15    VerticalAccuracyExt-r15    OPTIONAL
    ]]
}

```

850

```

ResponseTime ::= SEQUENCE {
    time
    ...,
    [[ responseTimeEarlyFix-r12 INTEGER (1..128)          OPTIONAL
    ]],
    [[ unit-r15                 ENUMERATED { ten-seconds, ... } OPTIONAL
    ]]
}

```

FIG. 8

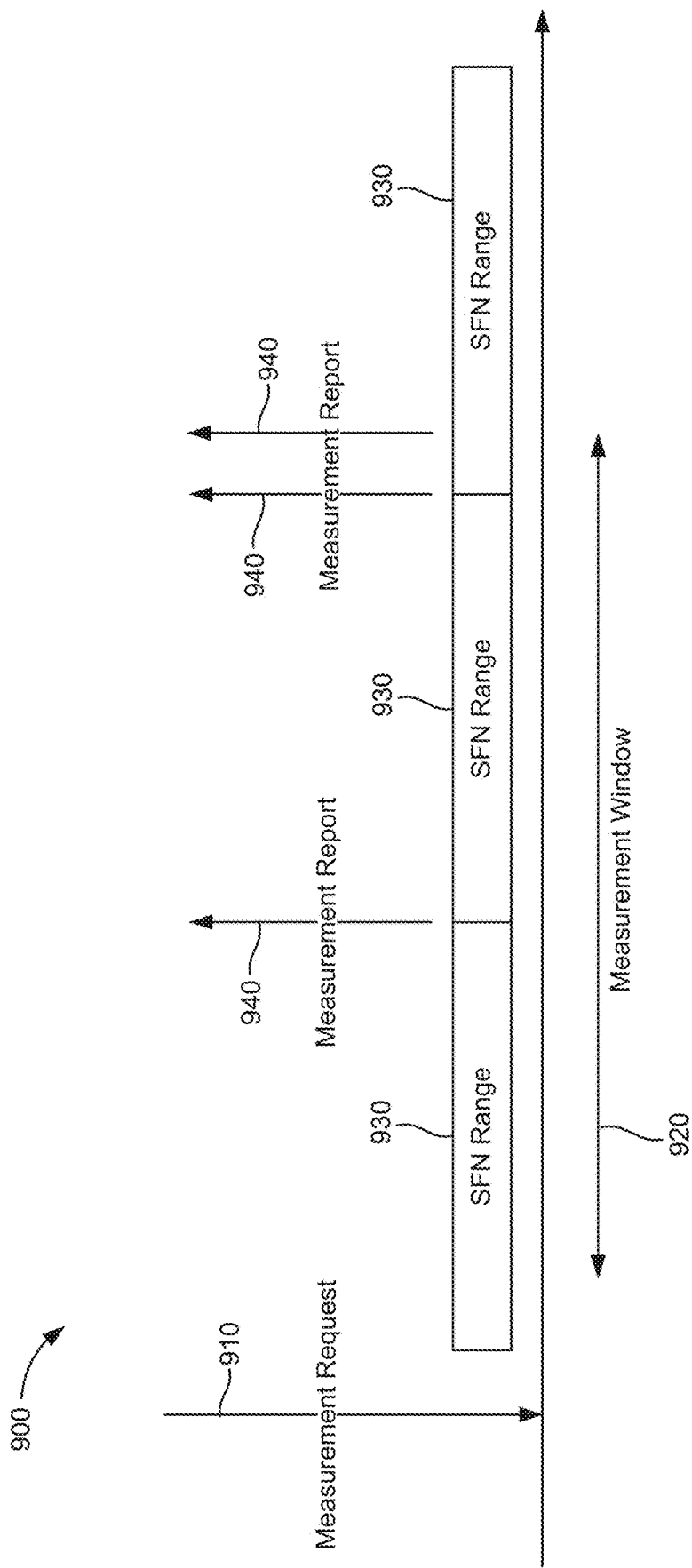
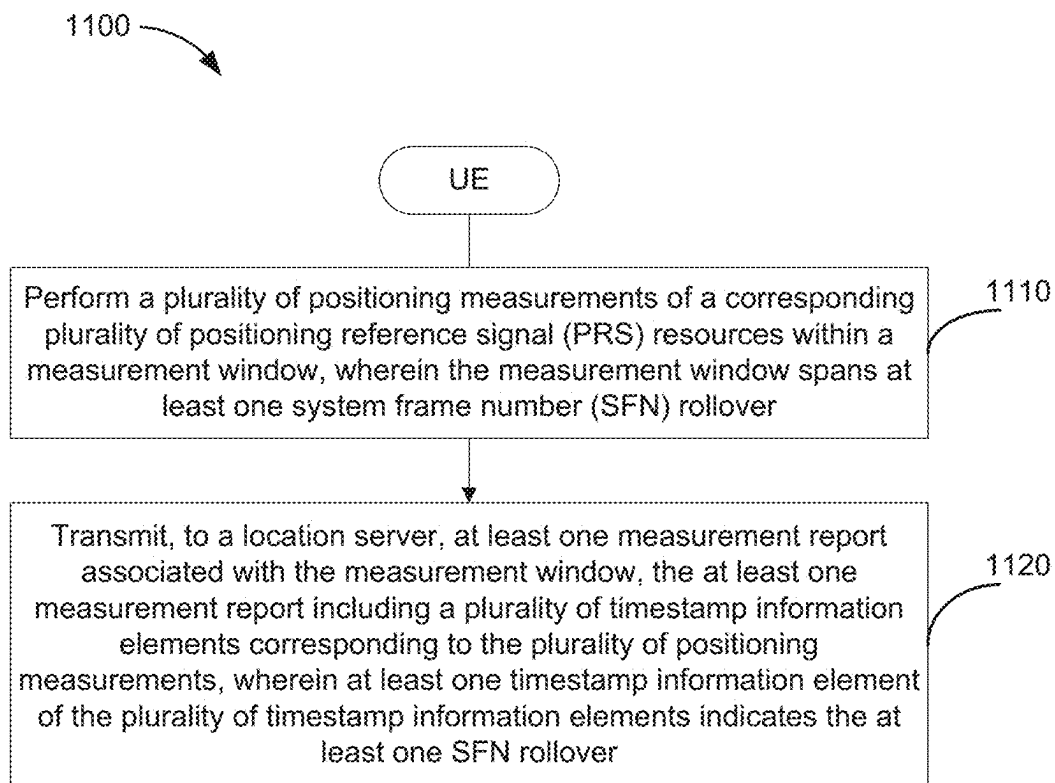


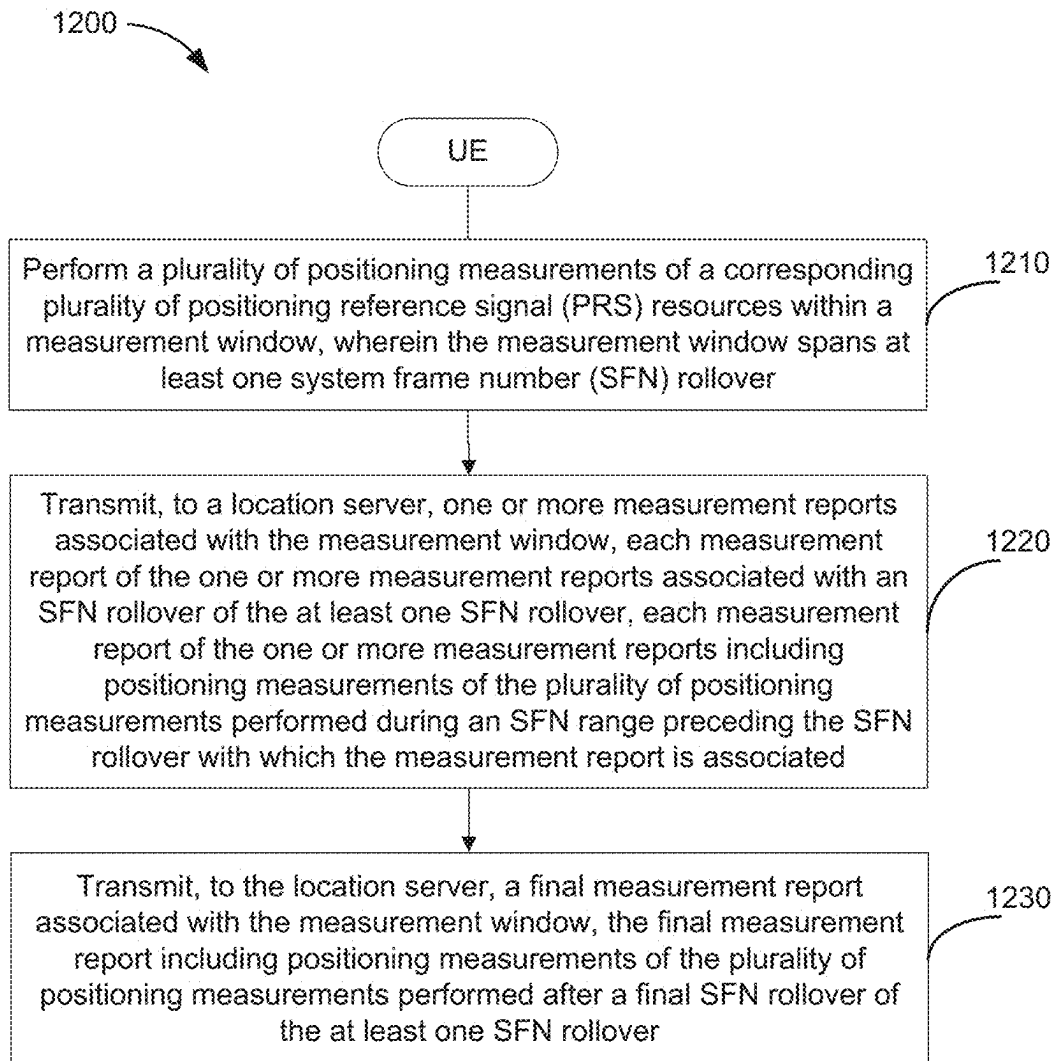
FIG. 9

1000 

```
NR-TimeStamp-r16 ::= SEQUENCE {
    dl-PRS-ID-r16          INTEGER (0..255),
    nr-PhysCellID-r16      NR-PhysCellID-r16
    nr-CellGlobalID-r16    NCGI-r15
    nr-ARFCN-r16           ARFCN-ValueNR-r15
    nr-SFN-r16             INTEGER (0..1023),
    nr-Slot-r16            CHOICE {
        scs15-r16          INTEGER (0..9),
        scs30-r16          INTEGER (0..19),
        scs60-r16          INTEGER (0..39),
        scs120-r16         INTEGER (0..79)
    },
    ...
    [ Nr-HSFN-r17          INTEGER (0..1023), OPTIONAL, -- Need ON ]
}
```

FIG. 10

**FIG. 11**

**FIG. 12**

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MEASUREMENT REPORTING ENHANCEMENTS IN BATCH MODE REPORTING

CROSS-REFERENCE TO RELATED APPLICATIONS

The present Application for Patent claims the benefit of U.S. Provisional application Ser. No. 17/405,495, entitled “MEASUREMENT REPORTING ENHANCEMENTS IN BATCH MODE REPORTING,” filed Aug. 18, 2021, assigned to the assignee hereof, and expressly incorporated herein by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

Aspects of the disclosure relate generally to wireless communications.

2. Description of the Related Art

Wireless communication systems have developed through various generations, including a first-generation analog wireless phone service (1G), a second-generation (2G) digital wireless phone service (including interim 2.5G and 2.75G networks), a third-generation (3G) high speed data, Internet-capable wireless service and a fourth-generation (4G) service (e.g., Long Term Evolution (LTE) or WiMax). There are presently many different types of wireless communication systems in use, including cellular and personal communications service (PCS) systems. Examples of known cellular systems include the cellular analog advanced mobile phone system (AMPS), and digital cellular systems based on code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), the Global System for Mobile communications (GSM), etc.

A fifth generation (5G) wireless standard, referred to as New Radio (NR), calls for higher data transfer speeds, greater numbers of connections, and better coverage, among other improvements. The 5G standard, according to the Next Generation Mobile Networks Alliance, is designed to provide data rates of several tens of megabits per second to each of tens of thousands of users, with 1 gigabit per second to tens of workers on an office floor. Several hundreds of thousands of simultaneous connections should be supported in order to support large sensor deployments. Consequently, the spectral efficiency of 5G mobile communications should be significantly enhanced compared to the current 4G standard. Furthermore, signaling efficiencies should be enhanced and latency should be substantially reduced compared to current standards.

SUMMARY

The following presents a simplified summary relating to one or more aspects disclosed herein. Thus, the following summary should not be considered an extensive overview relating to all contemplated aspects, nor should the following summary be considered to identify key or critical elements relating to all contemplated aspects or to delineate the scope associated with any particular aspect. Accordingly, the following summary has the sole purpose to present certain concepts relating to one or more aspects relating to

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the mechanisms disclosed herein in a simplified form to precede the detailed description presented below.

In an aspect, a method of wireless positioning performed by a user equipment (UE) includes performing a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmitting, to a location server, at least one measurement report associated with the measurement window, the at least one measurement report including a plurality of timestamp information elements corresponding to the plurality of positioning measurements, wherein at least one timestamp information element of the plurality of timestamp information elements indicates the at least one SFN rollover.

In an aspect, a method of wireless positioning performed by a user equipment (UE) includes performing a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmitting, to a location server, one or more measurement reports associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range preceding the SFN rollover with which the measurement report is associated; and transmitting, to the location server, a final measurement report associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning measurements performed after a final SFN rollover of the at least one SFN rollover.

In an aspect, a user equipment (UE) includes a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to: perform a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmit, via the at least one transceiver, to a location server, at least one measurement report associated with the measurement window, the at least one measurement report including a plurality of timestamp information elements corresponding to the plurality of positioning measurements, wherein at least one timestamp information element of the plurality of timestamp information elements indicates the at least one SFN rollover.

In an aspect, a user equipment (UE) includes a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to: perform a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmit, via the at least one transceiver, to a location server, one or more measurement reports associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range pre-

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ceding the SFN rollover with which the measurement report is associated; and transmit, via the at least one transceiver, to the location server, a final measurement report associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning measurements performed after a final SFN rollover of the at least one SFN rollover.

In an aspect, a user equipment (UE) includes means for performing a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and means for transmitting, to a location server, at least one measurement report associated with the measurement window, the at least one measurement report including a plurality of timestamp information elements corresponding to the plurality of positioning measurements, wherein at least one timestamp information element of the plurality of timestamp information elements indicates the at least one SFN rollover.

In an aspect, a user equipment (UE) includes means for performing a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and means for transmitting, to a location server, one or more measurement reports associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range preceding the SFN rollover with which the measurement report is associated; and means for transmitting, to the location server, a final measurement report associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning measurements performed after a final SFN rollover of the at least one SFN rollover.

In an aspect, a non-transitory computer-readable medium stores computer-executable instructions that, when executed by a user equipment (UE), cause the UE to: perform a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmit, to a location server, at least one measurement report associated with the measurement window, the at least one measurement report including a plurality of timestamp information elements corresponding to the plurality of positioning measurements, wherein at least one timestamp information element of the plurality of timestamp information elements indicates the at least one SFN rollover.

In an aspect, a non-transitory computer-readable medium stores computer-executable instructions that, when executed by a user equipment (UE), cause the UE to: perform a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmit, to a location server, one or more measurement reports associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or

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more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range preceding the SFN rollover with which the measurement report is associated; and transmit, to the location server, a final measurement report associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning measurements performed after a final SFN rollover of the at least one SFN rollover.

Other objects and advantages associated with the aspects disclosed herein will be apparent to those skilled in the art based on the accompanying drawings and detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings are presented to aid in the description of various aspects of the disclosure and are provided solely for illustration of the aspects and not limitation thereof.

FIG. 1 illustrates an example wireless communications system, according to aspects of the disclosure.

FIGS. 2A and 2B illustrate example wireless network structures, according to aspects of the disclosure.

FIGS. 3A, 3B, and 3C are simplified block diagrams of several sample aspects of components that may be employed in a user equipment (UE), a base station, and a network entity, respectively, and configured to support communications as taught herein.

FIG. 4 illustrates an example Long-Term Evolution (LTE) positioning protocol (LPP) call flow between a UE and a location server for performing positioning operations.

FIG. 5 is a diagram illustrating an example frame structure, according to aspects of the disclosure.

FIG. 6 is a diagram illustrating an example timeline for measurement reporting, according to aspects of the disclosure.

FIG. 7 illustrates an example “NR-TimeStamp” information element (IE) that defines the timestamp associated with a UE positioning reference signal (PRS) measurement.

FIG. 8 illustrates an example “QoS” IE and an example “ResponseTime” IE.

FIG. 9 is a diagram illustrating an example of measurement reporting when a measurement window spans multiple system frame number (SFN) rollovers, according to aspects of the disclosure.

FIG. 10 illustrates an example “NR-TimeStamp” IE that defines the timestamp associated with a UE PRS measurement, according to aspects of the disclosure.

FIGS. 11 and 12 illustrate example methods of wireless positioning, according to aspects of the disclosure.

DETAILED DESCRIPTION

Aspects of the disclosure are provided in the following description and related drawings directed to various examples provided for illustration purposes. Alternate aspects may be devised without departing from the scope of the disclosure. Additionally, well-known elements of the disclosure will not be described in detail or will be omitted so as not to obscure the relevant details of the disclosure.

The words “exemplary” and/or “example” are used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” and/or “example” is not necessarily to be construed as preferred or advantageous over other aspects. Likewise, the term

“aspects of the disclosure” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation.

Those of skill in the art will appreciate that the information and signals described below may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description below may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof, depending in part on the particular application, in part on the desired design, in part on the corresponding technology, etc.

Further, many aspects are described in terms of sequences of actions to be performed by, for example, elements of a computing device. It will be recognized that various actions described herein can be performed by specific circuits (e.g., application specific integrated circuits (ASICs)), by program instructions being executed by one or more processors, or by a combination of both. Additionally, the sequence(s) of actions described herein can be considered to be embodied entirely within any form of non-transitory computer-readable storage medium having stored therein a corresponding set of computer instructions that, upon execution, would cause or instruct an associated processor of a device to perform the functionality described herein. Thus, the various aspects of the disclosure may be embodied in a number of different forms, all of which have been contemplated to be within the scope of the claimed subject matter. In addition, for each of the aspects described herein, the corresponding form of any such aspects may be described herein as, for example, “logic configured to” perform the described action.

As used herein, the terms “user equipment” (UE) and “base station” are not intended to be specific or otherwise limited to any particular radio access technology (RAT), unless otherwise noted. In general, a UE may be any wireless communication device (e.g., a mobile phone, router, tablet computer, laptop computer, consumer asset locating device, wearable (e.g., smartwatch, glasses, augmented reality (AR)/virtual reality (VR) headset, etc.), vehicle (e.g., automobile, motorcycle, bicycle, etc.), Internet of Things (IoT) device, etc.) used by a user to communicate over a wireless communications network. A UE may be mobile or may (e.g., at certain times) be stationary, and may communicate with a radio access network (RAN). As used herein, the term “UE” may be referred to interchangeably as an “access terminal” or “AT,” a “client device,” a “wireless device,” a “subscriber device,” a “subscriber terminal,” a “subscriber station,” a “user terminal” or “UT,” a “mobile device,” a “mobile terminal,” a “mobile station,” or variations thereof. Generally, UEs can communicate with a core network via a RAN, and through the core network the UEs can be connected with external networks such as the Internet and with other UEs. Of course, other mechanisms of connecting to the core network and/or the Internet are also possible for the UEs, such as over wired access networks, wireless local area network (WLAN) networks (e.g., based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 specification, etc.) and so on.

A base station may operate according to one of several RATs in communication with UEs depending on the network in which it is deployed, and may be alternatively referred to as an access point (AP), a network node, a NodeB, an evolved NodeB (CNB), a next generation eNB (ng-eNB), a New Radio (NR) Node B (also referred to as a gNB or gNodeB), etc. A base station may be used primarily to

support wireless access by UEs, including supporting data, voice, and/or signaling connections for the supported UEs. In some systems a base station may provide purely edge node signaling functions while in other systems it may provide additional control and/or network management functions. A communication link through which UEs can send signals to a base station is called an uplink (UL) channel (e.g., a reverse traffic channel, a reverse control channel, an access channel, etc.). A communication link through which the base station can send signals to UEs is called a downlink (DL) or forward link channel (e.g., a paging channel, a control channel, a broadcast channel, a forward traffic channel, etc.). As used herein the term traffic channel (TCH) can refer to either an uplink/reverse or downlink/forward traffic channel.

The term “base station” may refer to a single physical transmission-reception point (TRP) or to multiple physical TRPs that may or may not be co-located. For example, where the term “base station” refers to a single physical TRP, the physical TRP may be an antenna of the base station corresponding to a cell (or several cell sectors) of the base station. Where the term “base station” refers to multiple co-located physical TRPs, the physical TRPs may be an array of antennas (e.g., as in a multiple-input multiple-output (MIMO) system or where the base station employs beamforming) of the base station. Where the term “base station” refers to multiple non-co-located physical TRPs, the physical TRPs may be a distributed antenna system (DAS) (a network of spatially separated antennas connected to a common source via a transport medium) or a remote radio head (RRH) (a remote base station connected to a serving base station). Alternatively, the non-co-located physical TRPs may be the serving base station receiving the measurement report from the UE and a neighbor base station whose reference radio frequency (RF) signals the UE is measuring. Because a TRP is the point from which a base station transmits and receives wireless signals, as used herein, references to transmission from or reception at a base station are to be understood as referring to a particular TRP of the base station.

In some implementations that support positioning of UEs, a base station may not support wireless access by UEs (e.g., may not support data, voice, and/or signaling connections for UEs), but may instead transmit reference signals to UEs to be measured by the UEs, and/or may receive and measure signals transmitted by the UEs. Such a base station may be referred to as a positioning beacon (e.g., when transmitting signals to UEs) and/or as a location measurement unit (e.g., when receiving and measuring signals from UEs).

An “RF signal” comprises an electromagnetic wave of a given frequency that transports information through the space between a transmitter and a receiver. As used herein, a transmitter may transmit a single “RF signal” or multiple “RF signals” to a receiver. However, the receiver may receive multiple “RF signals” corresponding to each transmitted RF signal due to the propagation characteristics of RF signals through multipath channels. The same transmitted RF signal on different paths between the transmitter and receiver may be referred to as a “multipath” RF signal. As used herein, an RF signal may also be referred to as a “wireless signal” or simply a “signal” where it is clear from the context that the term “signal” refers to a wireless signal or an RF signal.

FIG. 1 illustrates an example wireless communications system 100, according to aspects of the disclosure. The wireless communications system 100 (which may also be referred to as a wireless wide area network (WWAN)) may

include various base stations **102** (labeled “BS”) and various UEs **104**. The base stations **102** may include macro cell base stations (high power cellular base stations) and/or small cell base stations (low power cellular base stations). In an aspect, the macro cell base stations may include eNBs and/or ng-eNBs where the wireless communications system **100** corresponds to an LTE network, or gNBs where the wireless communications system **100** corresponds to a NR network, or a combination of both, and the small cell base stations may include femtocells, picocells, microcells, etc.

The base stations **102** may collectively form a RAN and interface with a core network **170** (e.g., an evolved packet core (EPC) or a 5G core (5GC)) through backhaul links **122**, and through the core network **170** to one or more location servers **172** (e.g., a location management function (LMF) or a secure user plane location (SUPL) location platform (SLP)). The location server(s) **172** may be part of core network **170** or may be external to core network **170**. A location server **172** may be integrated with a base station **102**. A UE **104** may communicate with a location server **172** directly or indirectly. For example, a UE **104** may communicate with a location server **172** via the base station **102** that is currently serving that UE **104**. A UE **104** may also communicate with a location server **172** through another path, such as via an application server (not shown), via another network, such as via a wireless local area network (WLAN) access point (AP) (e.g., AP **150** described below), and so on. For signaling purposes, communication between a UE **104** and a location server **172** may be represented as an indirect connection (e.g., through the core network **170**, etc.) or a direct connection (e.g., as shown via direct connection **128**), with the intervening nodes (if any) omitted from a signaling diagram for clarity.

In addition to other functions, the base stations **102** may perform functions that relate to one or more of transferring user data, radio channel ciphering and deciphering, integrity protection, header compression, mobility control functions (e.g., handover, dual connectivity), inter-cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, RAN sharing, multimedia broadcast multicast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, and delivery of warning messages. The base stations **102** may communicate with each other directly or indirectly (e.g., through the EPC/5GC) over backhaul links **134**, which may be wired or wireless.

The base stations **102** may wirelessly communicate with the UEs **104**. Each of the base stations **102** may provide communication coverage for a respective geographic coverage area **110**. In an aspect, one or more cells may be supported by a base station **102** in each geographic coverage area **110**. A “cell” is a logical communication entity used for communication with a base station (e.g., over some frequency resource, referred to as a carrier frequency, component carrier, carrier, band, or the like), and may be associated with an identifier (e.g., a physical cell identifier (PCI), an enhanced cell identifier (ECI), a virtual cell identifier (VCI), a cell global identifier (CGI), etc.) for distinguishing cells operating via the same or a different carrier frequency. In some cases, different cells may be configured according to different protocol types (e.g., machine-type communication (MTC), narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB), or others) that may provide access for different types of UEs. Because a cell is supported by a specific base station, the term “cell” may refer to either or both of the logical communication entity and the base station that

supports it, depending on the context. In addition, because a TRP is typically the physical transmission point of a cell, the terms “cell” and “TRP” may be used interchangeably. In some cases, the term “cell” may also refer to a geographic coverage area of a base station (e.g., a sector), insofar as a carrier frequency can be detected and used for communication within some portion of geographic coverage areas **110**.

While neighboring macro cell base station **102** geographic coverage areas **110** may partially overlap (e.g., in a handover region), some of the geographic coverage areas **110** may be substantially overlapped by a larger geographic coverage area **110**. For example, a small cell base station **102'** (labeled “SC” for “small cell”) may have a geographic coverage area **110'** that substantially overlaps with the geographic coverage area **110** of one or more macro cell base stations **102**. A network that includes both small cell and macro cell base stations may be known as a heterogeneous network. A heterogeneous network may also include home eNBs (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG).

The communication links **120** between the base stations **102** and the UEs **104** may include uplink (also referred to as reverse link) transmissions from a UE **104** to a base station **102** and/or downlink (DL) (also referred to as forward link) transmissions from a base station **102** to a UE **104**. The communication links **120** may use MIMO antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links **120** may be through one or more carrier frequencies. Allocation of carriers may be asymmetric with respect to downlink and uplink (e.g., more or less carriers may be allocated for downlink than for uplink).

The wireless communications system **100** may further include a wireless local area network (WLAN) access point (AP) **150** in communication with WLAN stations (STAs) **152** via communication links **154** in an unlicensed frequency spectrum (e.g., 5 GHz). When communicating in an unlicensed frequency spectrum, the WLAN STAs **152** and/or the WLAN AP **150** may perform a clear channel assessment (CCA) or listen before talk (LBT) procedure prior to communicating in order to determine whether the channel is available.

The small cell base station **102'** may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell base station **102'** may employ LTE or NR technology and use the same 5 GHz unlicensed frequency spectrum as used by the WLAN AP **150**. The small cell base station **102'**, employing LTE/5G in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network. NR in unlicensed spectrum may be referred to as NR-U. LTE in an unlicensed spectrum may be referred to as LTE-U, licensed assisted access (LAA), or MulteFire.

The wireless communications system **100** may further include a millimeter wave (mmW) base station **180** that may operate in mmW frequencies and/or near mmW frequencies in communication with a UE **182**. Extremely high frequency (EHF) is part of the RF in the electromagnetic spectrum. EHF has a range of 30 GHz to 300 GHz and a wavelength between 1 millimeter and 10 millimeters. Radio waves in this band may be referred to as a millimeter wave. Near mmW may extend down to a frequency of 3 GHz with a wavelength of 100 millimeters. The super high frequency (SHF) band extends between 3 GHz and 30 GHz, also referred to as centimeter wave. Communications using the mmW/near mmW radio frequency band have high path loss and a relatively short range. The mmW base station **180** and

the UE 182 may utilize beamforming (transmit and/or receive) over a mmW communication link 184 to compensate for the extremely high path loss and short range. Further, it will be appreciated that in alternative configurations, one or more base stations 102 may also transmit using mmW or near mmW and beamforming. Accordingly, it will be appreciated that the foregoing illustrations are merely examples and should not be construed to limit the various aspects disclosed herein.

Transmit beamforming is a technique for focusing an RF signal in a specific direction. Traditionally, when a network node (e.g., a base station) broadcasts an RF signal, it broadcasts the signal in all directions (omni-directionally). With transmit beamforming, the network node determines where a given target device (e.g., a UE) is located (relative to the transmitting network node) and projects a stronger downlink RF signal in that specific direction, thereby providing a faster (in terms of data rate) and stronger RF signal for the receiving device(s). To change the directionality of the RF signal when transmitting, a network node can control the phase and relative amplitude of the RF signal at each of the one or more transmitters that are broadcasting the RF signal. For example, a network node may use an array of antennas (referred to as a “phased array” or an “antenna array”) that creates a beam of RF waves that can be “steered” to point in different directions, without actually moving the antennas. Specifically, the RF current from the transmitter is fed to the individual antennas with the correct phase relationship so that the radio waves from the separate antennas add together to increase the radiation in a desired direction, while cancelling to suppress radiation in undesired directions.

Transmit beams may be quasi-co-located, meaning that they appear to the receiver (e.g., a UE) as having the same parameters, regardless of whether or not the transmitting antennas of the network node themselves are physically co-located. In NR, there are four types of quasi-co-location (QCL) relations. Specifically, a QCL relation of a given type means that certain parameters about a second reference RF signal on a second beam can be derived from information about a source reference RF signal on a source beam. Thus, if the source reference RF signal is QCL Type A, the receiver can use the source reference RF signal to estimate the Doppler shift, Doppler spread, average delay, and delay spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type B, the receiver can use the source reference RF signal to estimate the Doppler shift and Doppler spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type C, the receiver can use the source reference RF signal to estimate the Doppler shift and average delay of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type D, the receiver can use the source reference RF signal to estimate the spatial receive parameter of a second reference RF signal transmitted on the same channel.

In receive beamforming, the receiver uses a receive beam to amplify RF signals detected on a given channel. For example, the receiver can increase the gain setting and/or adjust the phase setting of an array of antennas in a particular direction to amplify (e.g., to increase the gain level of) the RF signals received from that direction. Thus, when a receiver is said to beamform in a certain direction, it means the beam gain in that direction is high relative to the beam gain along other directions, or the beam gain in that direction is the highest compared to the beam gain in that direction of

all other receive beams available to the receiver. This results in a stronger received signal strength (e.g., reference signal received power (RSRP), reference signal received quality (RSRQ), signal-to-interference-plus-noise ratio (SINR), etc.) of the RF signals received from that direction.

Transmit and receive beams may be spatially related. A spatial relation means that parameters for a second beam (e.g., a transmit or receive beam) for a second reference signal can be derived from information about a first beam (e.g., a receive beam or a transmit beam) for a first reference signal. For example, a UE may use a particular receive beam to receive a reference downlink reference signal (e.g., synchronization signal block (SSB)) from a base station. The UE can then form a transmit beam for sending an uplink reference signal (e.g., sounding reference signal (SRS)) to that base station based on the parameters of the receive beam.

Note that a “downlink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is forming the downlink beam to transmit a reference signal to a UE, the downlink beam is a transmit beam. If the UE is forming the downlink beam, however, it is a receive beam to receive the downlink reference signal. Similarly, an “uplink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is forming the uplink beam, it is an uplink receive beam, and if a UE is forming the uplink beam, it is an uplink transmit beam.

The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). It should be understood that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “Sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.

The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

With the above aspects in mind, unless specifically stated otherwise, it should be understood that the term “sub-6 GHz” or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term “millimeter wave” or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band.

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In a multi-carrier system, such as 5G, one of the carrier frequencies is referred to as the “primary carrier” or “anchor carrier” or “primary serving cell” or “PCell,” and the remaining carrier frequencies are referred to as “secondary carriers” or “secondary serving cells” or “SCells.” In carrier aggregation, the anchor carrier is the carrier operating on the primary frequency (e.g., FR1) utilized by a UE **104/182** and the cell in which the UE **104/182** either performs the initial radio resource control (RRC) connection establishment procedure or initiates the RRC connection re-establishment procedure. The primary carrier carries all common and UE-specific control channels, and may be a carrier in a licensed frequency (however, this is not always the case). A secondary carrier is a carrier operating on a second frequency (e.g., FR2) that may be configured once the RRC connection is established between the UE **104** and the anchor carrier and that may be used to provide additional radio resources. In some cases, the secondary carrier may be a carrier in an unlicensed frequency. The secondary carrier may contain only necessary signaling information and signals, for example, those that are UE-specific may not be present in the secondary carrier, since both primary uplink and downlink carriers are typically UE-specific. This means that different UEs **104/182** in a cell may have different downlink primary carriers. The same is true for the uplink primary carriers. The network is able to change the primary carrier of any UE **104/182** at any time. This is done, for example, to balance the load on different carriers. Because a “serving cell” (whether a PCell or an SCell) corresponds to a carrier frequency/component carrier over which some base station is communicating, the term “cell,” “serving cell,” “component carrier,” “carrier frequency,” and the like can be used interchangeably.

For example, still referring to FIG. 1, one of the frequencies utilized by the macro cell base stations **102** may be an anchor carrier (or “PCell”) and other frequencies utilized by the macro cell base stations **102** and/or the mmW base station **180** may be secondary carriers (“SCells”). The simultaneous transmission and/or reception of multiple carriers enables the UE **104/182** to significantly increase its data transmission and/or reception rates. For example, two 20 MHz aggregated carriers in a multi-carrier system would theoretically lead to a two-fold increase in data rate (i.e., 40 MHz), compared to that attained by a single 20 MHz carrier.

The wireless communications system **100** may further include a UE **164** that may communicate with a macro cell base station **102** over a communication link **120** and/or the mmW base station **180** over a mmW communication link **184**. For example, the macro cell base station **102** may support a PCell and one or more SCells for the UE **164** and the mmW base station **180** may support one or more SCells for the UE **164**.

In some cases, the UE **164** and the UE **182** may be capable of sidelink communication. Sidelink-capable UEs (SL-UEs) may communicate with base stations **102** over communication links **120** using the Uu interface (i.e., the air interface between a UE and a base station). SL-UEs (e.g., UE **164**, UE **182**) may also communicate directly with each other over a wireless sidelink **160** using the PC5 interface (i.e., the air interface between sidelink-capable UEs). A wireless sidelink (or just “sidelink”) is an adaptation of the core cellular (e.g., LTE, NR) standard that allows direct communication between two or more UEs without the communication needing to go through a base station. Sidelink communication may be unicast or multicast, and may be used for device-to-device (D2D) media-sharing, vehicle-to-vehicle (V2V) communication, vehicle-to-everything (V2X) com-

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munication (e.g., cellular V2X (cV2X) communication, enhanced V2X (eV2X) communication, etc.), emergency rescue applications, etc. One or more of a group of SL-UEs utilizing sidelink communications may be within the geographic coverage area **110** of a base station **102**. Other SL-UEs in such a group may be outside the geographic coverage area **110** of a base station **102** or be otherwise unable to receive transmissions from a base station **102**. In some cases, groups of SL-UEs communicating via sidelink communications may utilize a one-to-many (1:M) system in which each SL-UE transmits to every other SL-UE in the group. In some cases, a base station **102** facilitates the scheduling of resources for sidelink communications. In other cases, sidelink communications are carried out between SL-UEs without the involvement of a base station **102**.

In an aspect, the sidelink **160** may operate over a wireless communication medium of interest, which may be shared with other wireless communications between other vehicles and/or infrastructure access points, as well as other RATs. A “medium” may be composed of one or more time, frequency, and/or space communication resources (e.g., encompassing one or more channels across one or more carriers) associated with wireless communication between one or more transmitter/receiver pairs. In an aspect, the medium of interest may correspond to at least a portion of an unlicensed frequency band shared among various RATs. Although different licensed frequency bands have been reserved for certain communication systems (e.g., by a government entity such as the Federal Communications Commission (FCC) in the United States), these systems, in particular those employing small cell access points, have recently extended operation into unlicensed frequency bands such as the Unlicensed National Information Infrastructure (U-NII) band used by wireless local area network (WLAN) technologies, most notably IEEE 802.11x WLAN technologies generally referred to as “Wi-Fi.” Example systems of this type include different variants of CDMA systems, TDMA systems, FDMA systems, orthogonal FDMA (OFDMA) systems, single-carrier FDMA (SC-FDMA) systems, and so on.

Note that although FIG. 1 only illustrates two of the UEs as SL-UEs (i.e., UEs **164** and **182**), any of the illustrated UEs may be SL-UEs. Further, although only UE **182** was described as being capable of beamforming, any of the illustrated UEs, including UE **164**, may be capable of beamforming. Where SL-UEs are capable of beamforming, they may beamform towards each other (i.e., towards other SL-UEs), towards other UEs (e.g., UEs **104**), towards base stations (e.g., base stations **102**, **180**, small cell **102'**, access point **150**), etc. Thus, in some cases, UEs **164** and **182** may utilize beamforming over sidelink **160**.

In the example of FIG. 1, any of the illustrated UEs (shown in FIG. 1 as a single UE **104** for simplicity) may receive signals **124** from one or more Earth orbiting space vehicles (SVs) **112** (e.g., satellites). In an aspect, the SVs **112** may be part of a satellite positioning system that a UE **104** can use as an independent source of location information. A satellite positioning system typically includes a system of transmitters (e.g., SVs **112**) positioned to enable receivers (e.g., UEs **104**) to determine their location on or above the Earth based, at least in part, on positioning signals (e.g., signals **124**) received from the transmitters. Such a transmitter typically transmits a signal marked with a repeating pseudo-random noise (PN) code of a set number of chips. While typically located in SVs **112**, transmitters may sometimes be located on ground-based control stations, base

stations **102**, and/or other UEs **104**. A UE **104** may include one or more dedicated receivers specifically designed to receive signals **124** for deriving geo location information from the SVs **112**.

In a satellite positioning system, the use of signals **124** can be augmented by various satellite-based augmentation systems (SBAS) that may be associated with or otherwise enabled for use with one or more global and/or regional navigation satellite systems. For example an SBAS may include an augmentation system(s) that provides integrity information, differential corrections, etc., such as the Wide Area Augmentation System (WAAS), the European Geostationary Navigation Overlay Service (EGNOS), the Multifunctional Satellite Augmentation System (MSAS), the Global Positioning System (GPS) Aided Geo Augmented Navigation or GPS and Geo Augmented Navigation system (GAGAN), and/or the like. Thus, as used herein, a satellite positioning system may include any combination of one or more global and/or regional navigation satellites associated with such one or more satellite positioning systems.

In an aspect, SVs **112** may additionally or alternatively be part of one or more non-terrestrial networks (NTNs). In an NTN, an SV **112** is connected to an earth station (also referred to as a ground station, NTN gateway, or gateway), which in turn is connected to an element in a 5G network, such as a modified base station **102** (without a terrestrial antenna) or a network node in a 5GC. This element would in turn provide access to other elements in the 5G network and ultimately to entities external to the 5G network, such as Internet web servers and other user devices. In that way, a UE **104** may receive communication signals (e.g., signals **124**) from an SV **112** instead of, or in addition to, communication signals from a terrestrial base station **102**.

The wireless communications system **100** may further include one or more UEs, such as UE **190**, that connects indirectly to one or more communication networks via one or more device-to-device (D2D) peer-to-peer (P2P) links (referred to as "sidelinks"). In the example of FIG. 1, UE **190** has a D2D P2P link **192** with one of the UEs **104** connected to one of the base stations **102** (e.g., through which UE **190** may indirectly obtain cellular connectivity) and a D2D P2P link **194** with WLAN STA **152** connected to the WLAN AP **150** (through which UE **190** may indirectly obtain WLAN-based Internet connectivity). In an example, the D2D P2P links **192** and **194** may be supported with any well-known D2D RAT, such as LTE Direct (LTE-D), WiFi Direct (WiFi-D), Bluetooth®, and so on.

FIG. 2A illustrates an example wireless network structure **200**. For example, a 5GC **210** (also referred to as a Next Generation Core (NGC)) can be viewed functionally as control plane (C-plane) functions **214** (e.g., UE registration, authentication, network access, gateway selection, etc.) and user plane (U-plane) functions **212**, (e.g., UE gateway function, access to data networks, IP routing, etc.) which operate cooperatively to form the core network. User plane interface (NG-U) **213** and control plane interface (NG-C) **215** connect the gNB **222** to the 5GC **210** and specifically to the user plane functions **212** and control plane functions **214**, respectively. In an additional configuration, an ng-eNB **224** may also be connected to the 5GC **210** via NG-C **215** to the control plane functions **214** and NG-U **213** to user plane functions **212**. Further, ng-eNB **224** may directly communicate with gNB **222** via a backhaul connection **223**. In some configurations, a Next Generation RAN (NG-RAN) **220** may have one or more gNBs **222**, while other configurations include one or more of both ng-eNBs **224** and gNBs **222**.

Either (or both) gNB **222** or ng-eNB **224** may communicate with one or more UEs **204** (e.g., any of the UEs described herein).

Another optional aspect may include a location server **230**, which may be in communication with the 5GC **210** to provide location assistance for UE(s) **204**. The location server **230** can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The location server **230** can be configured to support one or more location services for UEs **204** that can connect to the location server **230** via the core network, 5GC **210**, and/or via the Internet (not illustrated). Further, the location server **230** may be integrated into a component of the core network, or alternatively may be external to the core network (e.g., a third party server, such as an original equipment manufacturer (OEM) server or service server).

FIG. 2B illustrates another example wireless network structure **250**. A 5GC **260** (which may correspond to 5GC **210** in FIG. 2A) can be viewed functionally as control plane functions, provided by an access and mobility management function (AMF) **264**, and user plane functions, provided by a user plane function (UPF) **262**, which operate cooperatively to form the core network (i.e., 5GC **260**). The functions of the AMF **264** include registration management, connection management, reachability management, mobility management, lawful interception, transport for session management (SM) messages between one or more UEs **204** (e.g., any of the UEs described herein) and a session management function (SMF) **266**, transparent proxy services for routing SM messages, access authentication and access authorization, transport for short message service (SMS) messages between the UE **204** and the short message service function (SMSF) (not shown), and security anchor functionality (SEAF). The AMF **264** also interacts with an authentication server function (AUSF) (not shown) and the UE **204**, and receives the intermediate key that was established as a result of the UE **204** authentication process. In the case of authentication based on a UMTS (universal mobile telecommunications system) subscriber identity module (USIM), the AMF **264** retrieves the security material from the AUSF. The functions of the AMF **264** also include security context management (SCM). The SCM receives a key from the SEAF that it uses to derive access-network specific keys. The functionality of the AMF **264** also includes location services management for regulatory services, transport for location services messages between the UE **204** and a location management function (LMF) **270** (which acts as a location server **230**), transport for location services messages between the NG-RAN **220** and the LMF **270**, evolved packet system (EPS) bearer identifier allocation for interworking with the EPS, and UE **204** mobility event notification. In addition, the AMF **264** also supports functionalities for non-3GPP (Third Generation Partnership Project) access networks.

Functions of the UPF **262** include acting as an anchor point for intra-/inter-RAT mobility (when applicable), acting as an external protocol data unit (PDU) session point of interconnect to a data network (not shown), providing packet routing and forwarding, packet inspection, user plane policy rule enforcement (e.g., gating, redirection, traffic steering), lawful interception (user plane collection), traffic usage reporting, quality of service (QoS) handling for the user plane (e.g., uplink/downlink rate enforcement, reflective QoS marking in the downlink), uplink traffic verification

(service data flow (SDF) to QoS flow mapping), transport level packet marking in the uplink and downlink, downlink packet buffering and downlink data notification triggering, and sending and forwarding of one or more “end markers” to the source RAN node. The UPF 262 may also support transfer of location services messages over a user plane between the UE 204 and a location server, such as an SLP 272.

The functions of the SMF 266 include session management, UE Internet protocol (IP) address allocation and management, selection and control of user plane functions, configuration of traffic steering at the UPF 262 to route traffic to the proper destination, control of part of policy enforcement and QoS, and downlink data notification. The interface over which the SMF 266 communicates with the AMF 264 is referred to as the N11 interface.

Another optional aspect may include an LMF 270, which may be in communication with the 5GC 260 to provide location assistance for UEs 204. The LMF 270 can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The LMF 270 can be configured to support one or more location services for UEs 204 that can connect to the LMF 270 via the core network, 5GC 260, and/or via the Internet (not illustrated). The SLP 272 may support similar functions to the LMF 270, but whereas the LMF 270 may communicate with the AMF 264, NG-RAN 220, and UEs 204 over a control plane (e.g., using interfaces and protocols intended to convey signaling messages and not voice or data), the SLP 272 may communicate with UEs 204 and external clients (e.g., third-party server 274) over a user plane (e.g., using protocols intended to carry voice and/or data like the transmission control protocol (TCP) and/or IP).

Yet another optional aspect may include a third-party server 274, which may be in communication with the LMF 270, the SLP 272, the 5GC 260 (e.g., via the AMF 264 and/or the UPF 262), the NG-RAN 220, and/or the UE 204 to obtain location information (e.g., a location estimate) for the UE 204. As such, in some cases, the third-party server 274 may be referred to as a location services (LCS) client or an external client. The third-party server 274 can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server.

User plane interface 263 and control plane interface 265 connect the 5GC 260, and specifically the UPF 262 and AMF 264, respectively, to one or more gNBs 222 and/or ng-eNBs 224 in the NG-RAN 220. The interface between gNB(s) 222 and/or ng-eNB(s) 224 and the AMF 264 is referred to as the “N2” interface, and the interface between gNB(s) 222 and/or ng-eNB(s) 224 and the UPF 262 is referred to as the “N3” interface. The gNB(s) 222 and/or ng-eNB(s) 224 of the NG-RAN 220 may communicate directly with each other via backhaul connections 223, referred to as the “Xn-C” interface. One or more of gNBs 222 and/or ng-eNBs 224 may communicate with one or more UEs 204 over a wireless interface, referred to as the “Uu” interface.

The functionality of a gNB 222 may be divided between a gNB central unit (gNB-CU) 226, one or more gNB distributed units (gNB-DUs) 228, and one or more gNB radio units (gNB-RUs) 229. A gNB-CU 226 is a logical node that includes the base station functions of transferring user

data, mobility control, radio access network sharing, positioning, session management, and the like, except for those functions allocated exclusively to the gNB-DU(s) 228. More specifically, the gNB-CU 226 generally hosts the radio resource control (RRC), service data adaptation protocol (SDAP), and packet data convergence protocol (PDCP) protocols of the gNB 222. A gNB-DU 228 is a logical node that generally hosts the radio link control (RLC) and medium access control (MAC) layer of the gNB 222. Its operation is controlled by the gNB-CU 226. One gNB-DU 228 can support one or more cells, and one cell is supported by only one gNB-DU 228. The interface 232 between the gNB-CU 226 and the one or more gNB-DUs 228 is referred to as the “F1” interface. The physical (PHY) layer functionality of a gNB 222 is generally hosted by one or more standalone gNB-RUs 229 that perform functions such as power amplification and signal transmission/reception. The interface between a gNB-DU 228 and a gNB-RU 229 is referred to as the “Fx” interface. Thus, a UE 204 communicates with the gNB-CU 226 via the RRC, SDAP, and PDCP layers, with a gNB-DU 228 via the RLC and MAC layers, and with a gNB-RU 229 via the PHY layer.

FIGS. 3A, 3B, and 3C illustrate several example components (represented by corresponding blocks) that may be incorporated into a UE 302 (which may correspond to any of the UEs described herein), a base station 304 (which may correspond to any of the base stations described herein), and a network entity 306 (which may correspond to or embody any of the network functions described herein, including the location server 230 and the LMF 270, or alternatively may be independent from the NG-RAN 220 and/or 5GC 210/260 infrastructure depicted in FIGS. 2A and 2B, such as a private network) to support the file transmission operations as taught herein. It will be appreciated that these components may be implemented in different types of apparatuses in different implementations (e.g., in an ASIC, in a system-on-chip (SoC), etc.). The illustrated components may also be incorporated into other apparatuses in a communication system. For example, other apparatuses in a system may include components similar to those described to provide similar functionality. Also, a given apparatus may contain one or more of the components. For example, an apparatus may include multiple transceiver components that enable the apparatus to operate on multiple carriers and/or communicate via different technologies.

The UE 302 and the base station 304 each include one or more wireless wide area network (WWAN) transceivers 310 and 350, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means for tuning, means for refraining from transmitting, etc.) via one or more wireless communication networks (not shown), such as an NR network, an LTE network, a GSM network, and/or the like. The WWAN transceivers 310 and 350 may each be connected to one or more antennas 316 and 356, respectively, for communicating with other network nodes, such as other UEs, access points, base stations (e.g., eNBs, gNBs), etc., via at least one designated RAT (e.g., NR, LTE, GSM, etc.) over a wireless communication medium of interest (e.g., some set of time/frequency resources in a particular frequency spectrum). The WWAN transceivers 310 and 350 may be variously configured for transmitting and encoding signals 318 and 358 (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals 318 and 358 (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the WWAN transceivers 310

and **350** include one or more transmitters **314** and **354**, respectively, for transmitting and encoding signals **318** and **358**, respectively, and one or more receivers **312** and **352**, respectively, for receiving and decoding signals **318** and **358**, respectively.

The UE **302** and the base station **304** each also include, at least in some cases, one or more short-range wireless transceivers **320** and **360**, respectively. The short-range wireless transceivers **320** and **360** may be connected to one or more antennas **326** and **366**, respectively, and provide means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means for tuning, means for refraining from transmitting, etc.) with other network nodes, such as other UEs, access points, base stations, etc., via at least one designated RAT (e.g., WiFi, LTE-D, Bluetooth®, Zigbee®, Z-Wave®, PC5, dedicated short-range communications (DSRC), wireless access for vehicular environments (WAVE), near-field communication (NFC), etc.) over a wireless communication medium of interest. The short-range wireless transceivers **320** and **360** may be variously configured for transmitting and encoding signals **328** and **368** (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals **328** and **368** (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the short-range wireless transceivers **320** and **360** include one or more transmitters **324** and **364**, respectively, for transmitting and encoding signals **328** and **368**, respectively, and one or more receivers **322** and **362**, respectively, for receiving and decoding signals **328** and **368**, respectively. As specific examples, the short-range wireless transceivers **320** and **360** may be WiFi transceivers, Bluetooth® transceivers, Zigbee® and/or Z-Wave® transceivers, NFC transceivers, or vehicle-to-vehicle (V2V) and/or vehicle-to-everything (V2X) transceivers.

The UE **302** and the base station **304** also include, at least in some cases, satellite signal receivers **330** and **370**. The satellite signal receivers **330** and **370** may be connected to one or more antennas **336** and **376**, respectively, and may provide means for receiving and/or measuring satellite positioning/communication signals **338** and **378**, respectively. Where the satellite signal receivers **330** and **370** are satellite positioning system receivers, the satellite positioning/communication signals **338** and **378** may be global positioning system (GPS) signals, global navigation satellite system (GLONASS) signals, Galileo signals, Beidou signals, Indian Regional Navigation Satellite System (NAVIC), Quasi-Zenith Satellite System (QZSS), etc. Where the satellite signal receivers **330** and **370** are non-terrestrial network (NTN) receivers, the satellite positioning/communication signals **338** and **378** may be communication signals (e.g., carrying control and/or user data) originating from a 5G network. The satellite signal receivers **330** and **370** may comprise any suitable hardware and/or software for receiving and processing satellite positioning/communication signals **338** and **378**, respectively. The satellite signal receivers **330** and **370** may request information and operations as appropriate from the other systems, and, at least in some cases, perform calculations to determine locations of the UE **302** and the base station **304**, respectively, using measurements obtained by any suitable satellite positioning system algorithm.

The base station **304** and the network entity **306** each include one or more network transceivers **380** and **390**, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, etc.) with other network entities (e.g., other base stations **304**, other network

entities **306**). For example, the base station **304** may employ the one or more network transceivers **380** to communicate with other base stations **304** or network entities **306** over one or more wired or wireless backhaul links. As another example, the network entity **306** may employ the one or more network transceivers **390** to communicate with one or more base station **304** over one or more wired or wireless backhaul links, or with other network entities **306** over one or more wired or wireless core network interfaces.

A transceiver may be configured to communicate over a wired or wireless link. A transceiver (whether a wired transceiver or a wireless transceiver) includes transmitter circuitry (e.g., transmitters **314**, **324**, **354**, **364**) and receiver circuitry (e.g., receivers **312**, **322**, **352**, **362**). A transceiver may be an integrated device (e.g., embodying transmitter circuitry and receiver circuitry in a single device) in some implementations, may comprise separate transmitter circuitry and separate receiver circuitry in some implementations, or may be embodied in other ways in other implementations. The transmitter circuitry and receiver circuitry of a wired transceiver (e.g., network transceivers **380** and **390** in some implementations) may be coupled to one or more wired network interface ports. Wireless transmitter circuitry (e.g., transmitters **314**, **324**, **354**, **364**) may include or be coupled to a plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such as an antenna array, that permits the respective apparatus (e.g., UE **302**, base station **304**) to perform transmit “beamforming,” as described herein. Similarly, wireless receiver circuitry (e.g., receivers **312**, **322**, **352**, **362**) may include or be coupled to a plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such as an antenna array, that permits the respective apparatus (e.g., UE **302**, base station **304**) to perform receive beamforming, as described herein. In an aspect, the transmitter circuitry and receiver circuitry may share the same plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such that the respective apparatus can only receive or transmit at a given time, not both at the same time. A wireless transceiver (e.g., WWAN transceivers **310** and **350**, short-range wireless transceivers **320** and **360**) may also include a network listen module (NLM) or the like for performing various measurements.

As used herein, the various wireless transceivers (e.g., transceivers **310**, **320**, **350**, and **360**, and network transceivers **380** and **390** in some implementations) and wired transceivers (e.g., network transceivers **380** and **390** in some implementations) may generally be characterized as “a transceiver,” “at least one transceiver,” or “one or more transceivers.” As such, whether a particular transceiver is a wired or wireless transceiver may be inferred from the type of communication performed. For example, backhaul communication between network devices or servers will generally relate to signaling via a wired transceiver, whereas wireless communication between a UE (e.g., UE **302**) and a base station (e.g., base station **304**) will generally relate to signaling via a wireless transceiver.

The UE **302**, the base station **304**, and the network entity **306** also include other components that may be used in conjunction with the operations as disclosed herein. The UE **302**, the base station **304**, and the network entity **306** include one or more processors **332**, **384**, and **394**, respectively, for providing functionality relating to, for example, wireless communication, and for providing other processing functionality. The processors **332**, **384**, and **394** may therefore provide means for processing, such as means for determining, means for calculating, means for receiving, means for transmitting, means for indicating, etc. In an aspect, the processors **332**, **384**, and **394** may include, for example, one

or more general purpose processors, multi-core processors, central processing units (CPUs), ASICs, digital signal processors (DSPs), field programmable gate arrays (FPGAs), other programmable logic devices or processing circuitry, or various combinations thereof.

The UE 302, the base station 304, and the network entity 306 include memory circuitry implementing memories 340, 386, and 396 (e.g., each including a memory device), respectively, for maintaining information (e.g., information indicative of reserved resources, thresholds, parameters, and so on). The memories 340, 386, and 396 may therefore provide means for storing, means for retrieving, means for maintaining, etc. In some cases, the UE 302, the base station 304, and the network entity 306 may include positioning component 342, 388, and 398, respectively. The positioning component 342, 388, and 398 may be hardware circuits that are part of or coupled to the processors 332, 384, and 394, respectively, that, when executed, cause the UE 302, the base station 304, and the network entity 306 to perform the functionality described herein. In other aspects, the positioning component 342, 388, and 398 may be external to the processors 332, 384, and 394 (e.g., part of a modem processing system, integrated with another processing system, etc.). Alternatively, the positioning component 342, 388, and 398 may be memory modules stored in the memories 340, 386, and 396, respectively, that, when executed by the processors 332, 384, and 394 (or a modem processing system, another processing system, etc.), cause the UE 302, the base station 304, and the network entity 306 to perform the functionality described herein. FIG. 3A illustrates possible locations of the positioning component 342, which may be, for example, part of the one or more WWAN transceivers 310, the memory 340, the one or more processors 332, or any combination thereof, or may be a standalone component. FIG. 3B illustrates possible locations of the positioning component 388, which may be, for example, part of the one or more WWAN transceivers 350, the memory 386, the one or more processors 384, or any combination thereof, or may be a standalone component. FIG. 3C illustrates possible locations of the positioning component 398, which may be, for example, part of the one or more network transceivers 390, the memory 396, the one or more processors 394, or any combination thereof, or may be a standalone component.

The UE 302 may include one or more sensors 344 coupled to the one or more processors 332 to provide means for sensing or detecting movement and/or orientation information that is independent of motion data derived from signals received by the one or more WWAN transceivers 310, the one or more short-range wireless transceivers 320, and/or the satellite signal receiver 330. By way of example, the sensor(s) 344 may include an accelerometer (e.g., a micro-electrical mechanical systems (MEMS) device), a gyroscope, a geomagnetic sensor (e.g., a compass), an altimeter (e.g., a barometric pressure altimeter), and/or any other type of movement detection sensor. Moreover, the sensor(s) 344 may include a plurality of different types of devices and combine their outputs in order to provide motion information. For example, the sensor(s) 344 may use a combination of a multi-axis accelerometer and orientation sensors to provide the ability to compute positions in two-dimensional (2D) and/or three-dimensional (3D) coordinate systems.

In addition, the UE 302 includes a user interface 346 providing means for providing indications (e.g., audible and/or visual indications) to a user and/or for receiving user input (e.g., upon user actuation of a sensing device such as a keypad, a touch screen, a microphone, and so on). Although

not shown, the base station 304 and the network entity 306 may also include user interfaces.

Referring to the one or more processors 384 in more detail, in the downlink, IP packets from the network entity 306 may be provided to the processor 384. The one or more processors 384 may implement functionality for an RRC layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The one or more processors 384 may provide RRC layer functionality associated with broadcasting of system information (e.g., master information block (MIB), system information blocks (SIBs)), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter-RAT mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer PDUs, error correction through automatic repeat request (ARQ), concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, scheduling information reporting, error correction, priority handling, and logical channel prioritization.

The transmitter 354 and the receiver 352 may implement Layer-1 (L1) functionality associated with various signal processing functions. Layer-1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The transmitter 354 handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an orthogonal frequency division multiplexing (OFDM) subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an inverse fast Fourier transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM symbol stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE 302. Each spatial stream may then be provided to one or more different antennas 356. The transmitter 354 may modulate an RF carrier with a respective spatial stream for transmission.

At the UE 302, the receiver 312 receives a signal through its respective antenna(s) 316. The receiver 312 recovers information modulated onto an RF carrier and provides the information to the one or more processors 332. The transmitter 314 and the receiver 312 implement Layer-1 functionality associated with various signal processing functions. The receiver 312 may perform spatial processing on the information to recover any spatial streams destined for the UE 302. If multiple spatial streams are destined for the UE 302, they may be combined by the receiver 312 into a single

OFDM symbol stream. The receiver 312 then converts the OFDM symbol stream from the time-domain to the frequency domain using a fast Fourier transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station 304. These soft decisions may be based on channel estimates computed by a channel estimator. The soft decisions are then decoded and de-interleaved to recover the data and control signals that were originally transmitted by the base station 304 on the physical channel. The data and control signals are then provided to the one or more processors 332, which implements Layer-3 (L3) and Layer-2 (L2) functionality.

In the uplink, the one or more processors 332 provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets from the core network. The one or more processors 332 are also responsible for error detection.

Similar to the functionality described in connection with the downlink transmission by the base station 304, the one or more processors 332 provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through hybrid automatic repeat request (HARQ), priority handling, and logical channel prioritization.

Channel estimates derived by the channel estimator from a reference signal or feedback transmitted by the base station 304 may be used by the transmitter 314 to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the transmitter 314 may be provided to different antenna(s) 316. The transmitter 314 may modulate an RF carrier with a respective spatial stream for transmission.

The uplink transmission is processed at the base station 304 in a manner similar to that described in connection with the receiver function at the UE 302. The receiver 352 receives a signal through its respective antenna(s) 356. The receiver 352 recovers information modulated onto an RF carrier and provides the information to the one or more processors 384.

In the uplink, the one or more processors 384 provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets from the UE 302. IP packets from the one or more processors 384 may be provided to the core network. The one or more processors 384 are also responsible for error detection.

For convenience, the UE 302, the base station 304, and/or the network entity 306 are shown in FIGS. 3A, 3B, and 3C as including various components that may be configured according to the various examples described herein. It will be appreciated, however, that the illustrated components may have different functionality in different designs. In

particular, various components in FIGS. 3A to 3C are optional in alternative configurations and the various aspects include configurations that may vary due to design choice, costs, use of the device, or other considerations. For example, in case of FIG. 3A, a particular implementation of UE 302 may omit the WWAN transceiver(s) 310 (e.g., a wearable device or tablet computer or PC or laptop may have Wi-Fi and/or Bluetooth capability without cellular capability), or may omit the short-range wireless transceiver(s) 320 (e.g., cellular-only, etc.), or may omit the satellite signal receiver 330, or may omit the sensor(s) 344, and so on. In another example, in case of FIG. 3B, a particular implementation of the base station 304 may omit the WWAN transceiver(s) 350 (e.g., a Wi-Fi "hotspot" access point without cellular capability), or may omit the short-range wireless transceiver(s) 360 (e.g., cellular-only, etc.), or may omit the satellite receiver 370, and so on. For brevity, illustration of the various alternative configurations is not provided herein, but would be readily understandable to one skilled in the art.

The various components of the UE 302, the base station 304, and the network entity 306 may be communicatively coupled to each other over data buses 334, 382, and 392, respectively. In an aspect, the data buses 334, 382, and 392 may form, or be part of, a communication interface of the UE 302, the base station 304, and the network entity 306, respectively. For example, where different logical entities are embodied in the same device (e.g., gNB and location server functionality incorporated into the same base station 304), the data buses 334, 382, and 392 may provide communication between them.

The components of FIGS. 3A, 3B, and 3C may be implemented in various ways. In some implementations, the components of FIGS. 3A, 3B, and 3C may be implemented in one or more circuits such as, for example, one or more processors and/or one or more ASICs (which may include one or more processors). Here, each circuit may use and/or incorporate at least one memory component for storing information or executable code used by the circuit to provide this functionality. For example, some or all of the functionality represented by blocks 310 to 346 may be implemented by processor and memory component(s) of the UE 302 (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Similarly, some or all of the functionality represented by blocks 350 to 388 may be implemented by processor and memory component(s) of the base station 304 (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Also, some or all of the functionality represented by blocks 390 to 398 may be implemented by processor and memory component(s) of the network entity 306 (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). For simplicity, various operations, acts, and/or functions are described herein as being performed "by a UE," "by a base station," "by a network entity," etc. However, as will be appreciated, such operations, acts, and/or functions may actually be performed by specific components or combinations of components of the UE 302, base station 304, network entity 306, etc., such as the processors 332, 384, 394, the transceivers 310, 320, 350, and 360, the memories 340, 386, and 396, the positioning component 342, 388, and 398, etc.

In some designs, the network entity 306 may be implemented as a core network component. In other designs, the network entity 306 may be distinct from a network operator or operation of the cellular network infrastructure (e.g., NG RAN 220 and/or 5GC 210/260). For example, the network

entity 306 may be a component of a private network that may be configured to communicate with the UE 302 via the base station 304 or independently from the base station 304 (e.g., over a non-cellular communication link, such as WiFi).

NR supports a number of cellular network-based positioning technologies, including downlink-based, uplink-based, and downlink-and-uplink-based positioning methods. Downlink-based positioning methods include observed time difference of arrival (OTDOA) in LTE, downlink time difference of arrival (DL-TDOA) in NR, and downlink angle-of-departure (DL-AoD) in NR. In an OTDOA or DL-TDOA positioning procedure, a UE measures the differences between the times of arrival (ToAs) of reference signals (e.g., positioning reference signals (PRS)) received from pairs of base stations, referred to as reference signal time difference (RSTD) or time difference of arrival (TDOA) measurements, and reports them to a positioning entity. More specifically, the UE receives the identifiers (IDs) of a reference base station (e.g., a serving base station) and multiple non-reference base stations in assistance data. The UE then measures the RSTD between the reference base station and each of the non-reference base stations. Based on the known locations of the involved base stations and the RSTD measurements, the positioning entity (e.g., the UE for UE-based positioning or a location server for UE-assisted positioning) can estimate the UE's location.

For DL-AoD positioning, the positioning entity uses a beam report from the UE of received signal strength measurements of multiple downlink transmit beams to determine the angle(s) between the UE and the transmitting base station(s). The positioning entity can then estimate the location of the UE based on the determined angle(s) and the known location(s) of the transmitting base station(s).

Uplink-based positioning methods include uplink time difference of arrival (UL-TDOA) and uplink angle-of-arrival (UL-AoA). UL-TDOA is similar to DL-TDOA, but is based on uplink reference signals (e.g., sounding reference signals (SRS)) transmitted by the UE. For UL-AoA positioning, one or more base stations measure the received signal strength of one or more uplink reference signals (e.g., SRS) received from a UE on one or more uplink receive beams. The positioning entity uses the signal strength measurements and the angle(s) of the receive beam(s) to determine the angle(s) between the UE and the base station(s). Based on the determined angle(s) and the known location(s) of the base station(s), the positioning entity can then estimate the location of the UE.

Downlink-and-uplink-based positioning methods include enhanced cell-ID (E-CID) positioning and multi-round-trip-time (RTT) positioning (also referred to as "multi-cell RTT" and "multi-RTT"). In an RTT procedure, a first entity (e.g., a base station or a UE) transmits a first RTT-related signal (e.g., a PRS or SRS) to a second entity (e.g., a UE or base station), which transmits a second RTT-related signal (e.g., an SRS or PRS) back to the first entity. Each entity measures the time difference between the time of arrival (ToA) of the received RTT-related signal and the transmission time of the transmitted RTT-related signal. This time difference is referred to as a reception-to-transmission (Rx-Tx) time difference. The Rx-Tx time difference measurement may be made, or may be adjusted, to include only a time difference between nearest subframe boundaries for the received and transmitted signals. Both entities may then send their Rx-Tx time difference measurement to a location server (e.g., an LMF 270), which calculates the round trip propagation time (i.e., RTT) between the two entities from the two Rx-Tx time difference measurements (e.g., as the sum of the two Rx-Tx

time difference measurements). Alternatively, one entity may send its Rx-Tx time difference measurement to the other entity, which then calculates the RTT. The distance between the two entities can be determined from the RTT and the known signal speed (e.g., the speed of light). For multi-RTT positioning, a first entity (e.g., a UE or base station) performs an RTT positioning procedure with multiple second entities (e.g., multiple base stations or UEs) to enable the location of the first entity to be determined (e.g., using multilateration) based on distances to, and the known locations of, the second entities. RTT and multi-RTT methods can be combined with other positioning techniques, such as UL-AoA and DL-AoD, to improve location accuracy.

The E-CID positioning method is based on radio resource management (RRM) measurements. In E-CID, the UE reports the serving cell ID, the timing advance (TA), and the identifiers, estimated timing, and signal strength of detected neighbor base stations. The location of the UE is then estimated based on this information and the known locations of the base station(s).

To assist positioning operations, a location server (e.g., location server 230, LMF 270, SLP 272) may provide assistance data to the UE. For example, the assistance data may include identifiers of the base stations (or the cells/TRPs of the base stations) from which to measure reference signals, the reference signal configuration parameters (e.g., the number of consecutive positioning subframes, periodicity of positioning subframes, muting sequence, frequency hopping sequence, reference signal identifier, reference signal bandwidth, etc.), and/or other parameters applicable to the particular positioning method. Alternatively, the assistance data may originate directly from the base stations themselves (e.g., in periodically broadcasted overhead messages, etc.). In some cases, the UE may be able to detect neighbor network nodes itself without the use of assistance data.

In the case of an OTDOA or DL-TDOA positioning procedure, the assistance data may further include an expected RSTD value and an associated uncertainty, or search window, around the expected RSTD. In some cases, the value range of the expected RSTD may be ± 500 microseconds (μ s). In some cases, when any of the resources used for the positioning measurement are in FR1, the value range for the uncertainty of the expected RSTD may be ± 32 μ s. In other cases, when all of the resources used for the positioning measurement(s) are in FR2, the value range for the uncertainty of the expected RSTD may be ± 8 μ s.

A location estimate may be referred to by other names, such as a position estimate, location, position, position fix, fix, or the like. A location estimate may be geodetic and comprise coordinates (e.g., latitude, longitude, and possibly altitude) or may be civic and comprise a street address, postal address, or some other verbal description of a location. A location estimate may further be defined relative to some other known location or defined in absolute terms (e.g., using latitude, longitude, and possibly altitude). A location estimate may include an expected error or uncertainty (e.g., by including an area or volume within which the location is expected to be included with some specified or default level of confidence).

FIG. 4 illustrates an example Long-Term Evolution (LTE) positioning protocol (LPP) procedure 400 between a UE 404 and a location server (illustrated as a location management function (LMF) 470) for performing positioning operations. As illustrated in FIG. 4, positioning of the UE 404 is supported via an exchange of LPP messages between the UE 404 and the LMF 470. The LPP messages may be exchanged

between UE 404 and the LMF 470 via the UE's 404 serving base station (illustrated as a serving gNB 402) and a core network (not shown). The LPP procedure 400 may be used to position the UE 404 in order to support various location-related services, such as navigation for UE 404 (or for the user of UE 404), or for routing, or for provision of an accurate location to a public safety answering point (PSAP) in association with an emergency call from UE 404 to a PSAP, or for some other reason. The LPP procedure 400 may also be referred to as a positioning session, and there may be multiple positioning sessions for different types of positioning methods (e.g., downlink time difference of arrival (DL-TDOA), round-trip-time (RTT), enhanced cell identity (E-CID), etc.).

Initially, the UE 404 may receive a request for its positioning capabilities from the LMF 470 at stage 410 (e.g., an LPP Request Capabilities message). At stage 420, the UE 404 provides its positioning capabilities to the LMF 470 relative to the LPP protocol by sending an LPP Provide Capabilities message to LMF 470 indicating the position methods and features of these position methods that are supported by the UE 404 using LPP. The capabilities indicated in the LPP Provide Capabilities message may, in some aspects, indicate the type of positioning the UE 404 supports (e.g., DL-TDOA, RTT, E-CID, etc.) and may indicate the capabilities of the UE 404 to support those types of positioning.

Upon reception of the LPP Provide Capabilities message, at stage 420, the LMF 470 determines to use a particular type of positioning method (e.g., DL-TDOA, RTT, E-CID, etc.) based on the indicated type(s) of positioning the UE 404 supports and determines a set of one or more transmission-reception points (TRPs) from which the UE 404 is to measure downlink positioning reference signals or towards which the UE 404 is to transmit uplink positioning reference signals. At stage 430, the LMF 470 sends an LPP Provide Assistance Data message to the UE 404 identifying the set of TRPs.

In some implementations, the LPP Provide Assistance Data message at stage 430 may be sent by the LMF 470 to the UE 404 in response to an LPP Request Assistance Data message sent by the UE 404 to the LMF 470 (not shown in FIG. 4). An LPP Request Assistance Data message may include an identifier of the UE's 404 serving TRP and a request for the positioning reference signal (PRS) configuration of neighboring TRPs.

At stage 440, the LMF 470 sends a request for location information to the UE 404. The request may be an LPP Request Location Information message. This message usually includes information elements defining the location information type, desired accuracy of the location estimate, and response time (i.e., desired latency). Note that a low latency requirement allows for a longer response time while a high latency requirement requires a shorter response time. However, a long response time is referred to as high latency and a short response time is referred to as low latency.

Note that in some implementations, the LPP Provide Assistance Data message sent at stage 430 may be sent after the LPP Request Location Information message at 440 if, for example, the UE 404 sends a request for assistance data to LMF 470 (e.g., in an LPP Request Assistance Data message, not shown in FIG. 4) after receiving the request for location information at stage 440.

At stage 450, the UE 404 utilizes the assistance information received at stage 430 and any additional data (e.g., a desired location accuracy or a maximum response time) received at stage 440 to perform positioning operations (e.g.,

measurements of DL-PRS, transmission of UL-PRS, etc.) for the selected positioning method.

At stage 460, the UE 404 may send an LPP Provide Location Information message to the LMF 470 conveying the results of any measurements that were obtained at stage 450 (e.g., time of arrival (ToA), reference signal time difference (RSTD), reception-to-transmission (Rx-Tx), etc.) and before or when any maximum response time has expired (e.g., a maximum response time provided by the LMF 470 at stage 440). The LPP Provide Location Information message at stage 460 may also include the time (or times) at which the positioning measurements were obtained and the identity of the TRP(s) from which the positioning measurements were obtained. Note that the time between the request for location information at 440 and the response at 460 is the "response time" and indicates the latency of the positioning session.

The LMF 470 computes an estimated location of the UE 404 using the appropriate positioning techniques (e.g., DL-TDOA, RTT, E-CID, etc.) based, at least in part, on measurements received in the LPP Provide Location Information message at stage 460.

Various frame structures may be used to support downlink and uplink transmissions between network nodes (e.g., base stations and UEs). FIG. 5 is a diagram 500 illustrating an example frame structure, according to aspects of the disclosure. The frame structure may be a downlink or uplink frame structure. Other wireless communications technologies may have different frame structures and/or different channels.

LTE, and in some cases NR, utilizes OFDM on the downlink and single-carrier frequency division multiplexing (SC-FDM) on the uplink. Unlike LTE, however, NR has an option to use OFDM on the uplink as well. OFDM and SC-FDM partition the system bandwidth into multiple (K) orthogonal subcarriers, which are also commonly referred to as tones, bins, etc. Each subcarrier may be modulated with data. In general, modulation symbols are sent in the frequency domain with OFDM and in the time domain with SC-FDM. The spacing between adjacent subcarriers may be fixed, and the total number of subcarriers (K) may be dependent on the system bandwidth. For example, the spacing of the subcarriers may be 15 kilohertz (kHz) and the minimum resource allocation (resource block) may be 12 subcarriers (or 180 kHz). Consequently, the nominal FFT size may be equal to 128, 256, 512, 1024, or 2048 for system bandwidth of 1.25, 2.5, 5, 10, or 20 megahertz (MHz), respectively. The system bandwidth may also be partitioned into subbands. For example, a subband may cover 1.08 MHz (i.e., 6 resource blocks), and there may be 1, 2, 4, 8, or 16 subbands for system bandwidth of 1.25, 2.5, 5, 10, or 20 MHz, respectively.

LTE supports a single numerology (subcarrier spacing (SCS), symbol length, etc.). In contrast, NR may support multiple numerologies (μ), for example, subcarrier spacings of 15 kHz ($\mu=0$), 30 kHz ($\mu=1$), 60 kHz ($\mu=2$), 120 kHz ($\mu=3$), and 240 kHz ($\mu=4$) or greater may be available. In each subcarrier spacing, there are 14 symbols per slot. For 15 kHz SCS ($\mu=0$), there is one slot per subframe, 10 slots per frame, the slot duration is 1 millisecond (ms), the symbol duration is 66.7 microseconds (μ s), and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 50. For 30 kHz SCS ($\mu=1$), there are two slots per subframe, 20 slots per frame, the slot duration is 0.5 ms, the symbol duration is 33.3 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 100. For 60 kHz SCS ($\mu=2$), there are four slots per subframe, 40 slots per frame, the slot duration is 0.25 ms, the symbol duration is

16.7 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 200. For 120 kHz SCS ($\mu=3$), there are eight slots per subframe, 80 slots per frame, the slot duration is 0.125 ms, the symbol duration is 8.33 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 400. For 240 kHz SCS ($\mu=4$), there are 16 slots per subframe, 160 slots per frame, the slot duration is 0.0625 ms, the symbol duration is 4.17 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 800.

In the example of FIG. 5, a numerology of 15 kHz is used. Thus, in the time domain, a 10 ms frame is divided into 10 equally sized subframes of 1 ms each, and each subframe includes one time slot. In FIG. 5, time is represented horizontally (on the X axis) with time increasing from left to right, while frequency is represented vertically (on the Y axis) with frequency increasing (or decreasing) from bottom to top.

A resource grid may be used to represent time slots, each time slot including one or more time-concurrent resource blocks (RBs) (also referred to as physical RBs (PRBs)) in the frequency domain. The resource grid is further divided into multiple resource elements (REs). An RE may correspond to one symbol length in the time domain and one subcarrier in the frequency domain. In the numerology of FIG. 5, for a normal cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and seven consecutive symbols in the time domain, for a total of 84 REs. For an extended cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and six consecutive symbols in the time domain, for a total of 72 REs. The number of bits carried by each RE depends on the modulation scheme.

Some of the REs may carry reference (pilot) signals (RS). The reference signals may include positioning reference signals (PRS), tracking reference signals (TRS), phase tracking reference signals (PTRS), cell-specific reference signals (CRS), channel state information reference signals (CSI-RS), demodulation reference signals (DMRS), primary synchronization signals (PSS), secondary synchronization signals (SSS), synchronization signal blocks (SSBs), sounding reference signals (SRS), etc., depending on whether the illustrated frame structure is used for uplink or downlink communication. FIG. 5 illustrates example locations of REs carrying a reference signal (labeled "R").

Downlink PRS (DL-PRS) have been defined for NR positioning to enable UEs to detect and measure more neighboring TRPs. Several configurations are supported to enable a variety of deployments (e.g., indoor, outdoor, sub-6 GHz, mmW). In addition, both UE-assisted (where a network entity estimates the location of a target UE) and UE-based (where the target UE estimates its own location) positioning are supported. The following table illustrates various types of reference signals that can be used for various positioning methods supported in NR.

TABLE 1

DL/UL Reference Signals	UE Measurements	To support the following positioning techniques
DL-PRS	DL-RSTD	DL-TDOA
DL-PRS	DL-PRS RSRP	DL-TDOA, DL-AoD, Multi-RTT
DL-PRS/SRS-for-positioning	UE Rx-Tx time difference	Multi-RTT
SSB/CSI-RS for	Synchronization Signal	E-CID

TABLE 1-continued

DL/UL Reference Signals	UE Measurements	To support the following positioning techniques
5 RRM	(SS)-RSRP (RSRP for RRM), SS-RSRQ (for RRM), CSI-RSRP (for RRM), CSI-RSRQ (for RRM)	

A collection of resource elements (REs) that are used for transmission of PRS is referred to as a "PRS resource." The collection of resource elements can span multiple PRBs in the frequency domain and 'N' (such as 1 or more) consecutive symbol(s) within a slot in the time domain. In a given OFDM symbol in the time domain, a PRS resource occupies consecutive PRBs in the frequency domain.

The transmission of a PRS resource within a given PRB has a particular comb size (also referred to as the "comb density"). A comb size 'N' represents the subcarrier spacing (or frequency/tone spacing) within each symbol of a PRS resource configuration.

Specifically, for a comb size 'N,' PRS are transmitted in every Nth subcarrier of a symbol of a PRB. For example, for comb-4, for each symbol of the PRS resource configuration, REs corresponding to every fourth subcarrier (such as subcarriers 0, 4, 8) are used to transmit PRS of the PRS resource. Currently, comb sizes of comb-2, comb-4, comb-6, and comb-12 are supported for DL-PRS. FIG. 5 illustrates an example PRS resource configuration for comb-4 (which spans four symbols). That is, the locations of the shaded REs (labeled "R") indicate a comb-4 PRS resource configuration.

Currently, a DL-PRS resource may span 2, 4, 6, or 12 consecutive symbols within a slot with a fully frequency-domain staggered pattern. A DL-PRS resource can be configured in any higher layer configured downlink or flexible (FL) symbol of a slot. There may be a constant energy per resource element (EPRE) for all REs of a given DL-PRS resource. The following are the frequency offsets from symbol to symbol for comb sizes 2, 4, 6, and 12 over 2, 4, 6, and 12 symbols. 2-symbol comb-2: {0, 1}; 4-symbol comb-2: {0, 1, 0, 1}; 6-symbol comb-2: {0, 1, 0, 1, 0, 1}; 12-symbol comb-2: {0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1}; 4-symbol comb-4: {0, 2, 1, 3} (as in the example of FIG. 5); 12-symbol comb-4: {0, 2, 1, 3, 0, 2, 1, 3, 0, 2, 1, 3}; 6-symbol comb-6: {0, 3, 1, 4, 2, 5}; 12-symbol comb-6: {0, 3, 1, 4, 2, 5, 0, 3, 1, 4, 2, 5}; and 12-symbol comb-12: {0, 6, 3, 9, 1, 7, 4, 10, 2, 8, 5, 11}.

A "PRS resource set" is a set of PRS resources used for the transmission of PRS signals, where each PRS resource has a PRS resource ID. In addition, the PRS resources in a PRS resource set are associated with the same TRP. A PRS resource set is identified by a PRS resource set ID and is associated with a particular TRP (identified by a TRP ID). In addition, the PRS resources in a PRS resource set have the same periodicity, a common muting pattern configuration, and the same repetition factor (such as "PRS-ResourceRepetitionFactor") across slots. The periodicity is the time from the first repetition of the first PRS resource of a first PRS instance to the same first repetition of the same first PRS resource of the next PRS instance. The periodicity may have a length selected from $2^{\mu} \cdot \{4, 5, 8, 10, 16, 20, 32, 40, 64, 80, 160, 320, 640, 1280, 2560, 5120, 10240\}$ slots, with $\mu=0, 1, 2, 3$. The repetition factor may have a length selected from $\{1, 2, 4, 6, 8, 16, 32\}$ slots.

A PRS resource ID in a PRS resource set is associated with a single beam (or beam ID) transmitted from a single

TRP (where a TRP may transmit one or more beams). That is, each PRS resource of a PRS resource set may be transmitted on a different beam, and as such, a “PRS resource,” or simply “resource,” also can be referred to as a “beam.” Note that this does not have any implications on whether the TRPs and the beams on which PRS are transmitted are known to the UE.

A “PRS instance” or “PRS occasion” is one instance of a periodically repeated time window (such as a group of one or more consecutive slots) where PRS are expected to be transmitted. A PRS occasion also may be referred to as a “PRS positioning occasion,” a “PRS positioning instance,” a “positioning occasion,” a “positioning instance,” a “positioning repetition,” or simply an “occasion,” an “instance,” or a “repetition.”

A “positioning frequency layer” (also referred to simply as a “frequency layer”) is a collection of one or more PRS resource sets across one or more TRPs that have the same values for certain parameters. Specifically, the collection of PRS resource sets has the same subcarrier spacing and cyclic prefix (CP) type (meaning all numerologies supported for the physical downlink shared channel (PDSCH) are also supported for PRS), the same Point A, the same value of the downlink PRS bandwidth, the same start PRB (and center frequency), and the same comb-size. The Point A parameter takes the value of the parameter “ARFCN-ValueNR” (where “ARFCN” stands for “absolute radio-frequency channel number”) and is an identifier/code that specifies a pair of physical radio channel used for transmission and reception. The downlink PRS bandwidth may have a granularity of four PRBs, with a minimum of 24 PRBs and a maximum of 272 PRBs. Currently, up to four frequency layers have been defined, and up to two PRS resource sets may be configured per TRP per frequency layer.

The concept of a frequency layer is somewhat like the concept of component carriers and bandwidth parts (BWPs), but different in that component carriers and BWPs are used by one base station (or a macro cell base station and a small cell base station) to transmit data channels, while frequency layers are used by several (usually three or more) base stations to transmit PRS. A UE may indicate the number of frequency layers it can support when it sends the network its positioning capabilities, such as during an LTE positioning protocol (LPP) session. For example, a UE may indicate whether it can support one or four positioning frequency layers.

Note that the terms “positioning reference signal” and “PRS” generally refer to specific reference signals that are used for positioning in NR and LTE systems. However, as used herein, the terms “positioning reference signal” and “PRS” may also refer to any type of reference signal that can be used for positioning, such as but not limited to, PRS as defined in LTE and NR, TRS, PTRS, CRS, CSI-RS, DMRS, PSS, SSS, SSB, SRS, UL-PRS, etc. In addition, the terms “positioning reference signal” and “PRS” may refer to downlink or uplink positioning reference signals, unless otherwise indicated by the context. If needed to further distinguish the type of PRS, a downlink positioning reference signal may be referred to as a “DL-PRS,” and an uplink positioning reference signal (e.g., an SRS-for-positioning, PTRS) may be referred to as an “UL-PRS.” In addition, for signals that may be transmitted in both the uplink and downlink (e.g., DMRS, PTRS), the signals may be prepended with “UL” or “DL” to distinguish the direction. For example, “UL-DMRS” may be differentiated from “DL-DMRS.”

A UE reports its capability to process PRS in the capability update (e.g., part of the UE positioning procedures with the UE **204** at stage **430b**). The assistance data received based on the UE’s capability information includes the information needed to perform measurements of PRS for the positioning session (e.g., the time and frequency configurations of the PRS resources to measure from one or more base stations/TRPs/cells). However, the assistance data may identify significantly more PRS resources to measure than the UE is capable of processing. For example, a UE may be able to process only up to five PRS resources, whereas the PRS assistance data may identify 20 PRS resources to measure.

Currently, in such cases, the UE selects the first five PRS resources for processing. Specifically, it has been agreed that when a UE is configured in the assistance data of a positioning method with a number of PRS resources beyond its capability, the UE assumes the PRS resources within a positioning frequency layer are sorted in a decreasing order of measurement priority. According to the current structure of the assistance data, the following priority is assumed: the 64 TRPs per frequency layer are sorted according to priority and the two PRS resource sets per TRP of the frequency layer are sorted according to priority. The four frequency layers may or may not be sorted according to priority and the 64 PRS resources of the PRS resource set per TRP per frequency layer may or may not be sorted according to priority. Note that the reference PRS resource indicated by the “nr-DL-PRS-ReferenceInfo-r16” LPP information element for each frequency layer has the highest priority, at least for DL-TDOA positioning procedures.

As such, the priority of PRS measurements (i.e., positioning measurements of PRS resources) is defined, but the priority of measurement reporting is not defined. Instead, it is up to UE implementation regarding how to report PRS measurement results in the measurement report.

FIG. **6** is a diagram **600** illustrating an example timeline for measurement reporting, according to aspects of the disclosure. In FIG. **6**, time is represented horizontally and signal strength is represented vertically. In the example of FIG. **6**, a UE is configured with a measurement periodicity T (e.g., 1 second), meaning the UE is configured to transmit a measurement report every T seconds. FIG. **6** illustrates a measurement periodicity T of 1 second between a first measurement report **610** and a second measurement report **630** (e.g., LPP Provide Location Information messages, as at stage **460**). In the example of FIG. **6**, within the measurement periodicity (or measurement report periodicity), there are six PRS occasions, or instances, **620**. In FIG. **6**, each PRS occasion **620** comprises multiple (e.g., six) PRS resources, as illustrated by the vertical arrows. The relative signal strength of these PRS resources is represented by the height of the vertical arrows. Each vertical arrow has a particular dash pattern, which identifies that same resource across PRS occasions **620**. That is, each PRS resource in a PRS occasion **620** is repeated across all PRS occasions **620**.

The UE may perform the same or different type of positioning measurement (e.g., ToA, RSRP, RSTD, etc.) of the PRS resources in each PRS occasion **620**. Because the UE is performing measurements of the same PRS resources across multiple PRS occasions **620**, the UE can report all of the measurements in a batch measurement report. That is, the measurement report **630** would be a batch measurement report. Within the measurement report **630**, there will be multiple measurements of each PRS resource with different timestamps. That is, every PRS measurement (i.e., every

positioning measurement of a PRS resource) in the measurement report **630** will have a timestamp to indicate the measurement time.

Currently, the timestamp associated with a measurement is reported in the LPP information element (IE) “NR-TimeStamp.” FIG. 7 illustrates an example “NR-TimeStamp” IE **700** that defines the timestamp associated with a UE PRS measurement. The following table describes the fields of the “NR-TimeStamp” IE **700**:

TABLE 2

NR-TimeStamp Field Descriptions	
dl-PRS-ID	This field specifies the DL-PRS ID of the TRP for which the parameter “nr-SFN” is applicable.
nr-ARFCN	This field specifies the ARFCN of the TRP associated with the “dl-PRS-ID.”
nr-SFN	This field specifies the NR system frame number (SFN) for the timestamp.
nr-Slot	This field specifies the NR slot number within the NR system frame number indicated by “nr-SFN” for the timestamp.

As shown in FIG. 7, the time span of “nr-SFN” is 0 to 1023 SFNs, which means it can take a value of up to 10.23 seconds. However, the maximum response time (e.g., the response time between stages **440** and **460** in FIG. 4) may be up to 128 seconds. The response time defines the measurement window for a positioning session and is defined in the LPP QoS IE, illustrated in FIG. 8. Specifically, FIG. 8 illustrates an example “QoS” IE **800** and an example “ResponseTime” IE **850**. As shown in FIG. 8, the “QoS” IE **800** includes a “responseTime” field that points to the “ResponseTime” IE **850**. Within the “ResponseTime” IE **850**, the “time” field indicates the maximum response time as measured between receipt of the LPP Request Location Information message (at stage **440** of FIG. 4) and transmission of an LPP Provide Location Information message (at stage **460** of FIG. 4). If the “unit” field is absent, this is given as an integer number of seconds between 1 and 128. If the “unit” field is present, the maximum response time is given in units of 10-seconds, between 10 and 1280 seconds.

The field “responseTimeEarlyFix” indicates the maximum response time as measured between receipt of the LPP Request Location Information message and transmission of an LPP Provide Location Information message containing early location measurements or an early location estimate. If the “unit” field is absent, this is given as an integer number of seconds between 1 and 128. If the “unit” field is present, the maximum response time is given in units of 10-seconds, between 10 and 1280 seconds. When this field is included, a target UE (the UE being positioned) should send an LPP Provide Location Information message (or more than one LPP Provide Location Information message if location information will not fit into a single message) containing early location information according to the “responseTimeEarlyFix” field and a subsequent LPP Provide Location Information message (or more than one LPP Provide Location Information message if location information will not fit into a single message) containing final location information according to the “time” field. A target UE will omit sending an LPP Provide Location Information message if the early location information is not available at the expiration of the time value in the “responseTimeEarlyFix” field. A location server should set the “responseTimeEarlyFix” field to a value less than that for the “time” field. A target UE should

ignore the “responseTimeEarlyFix” field if its value is not less than that for the “time” field.

The “unit” field indicates the unit of the “time” and “responseTimeEarlyFix” fields. The enumerated value “ten-seconds” corresponds to a resolution of 10 seconds. If this field is absent, the unit/resolution is 1 second.

In the case where the response time is greater than 10 seconds, meaning the measurement window is greater than 10 seconds, the SFN will rollover inside the measurement window. That is, the SFN will increment to a value of 1023 and then reset to 0. If an SFN rollover occurs during batch mode reporting, it will be very difficult for the location server to time-align all of the UE measurements (i.e., positioning measurements of DL-PRS and/or UL-PRS) with the corresponding gNB measurements (i.e., positioning measurements of UL-PRS and/or DL-PRS), making it difficult for the location server to align a positioning fix with a specific time in the past. Any misalignment between the UE and gNB measurements can therefore result in an error in the location estimate(s).

Accordingly, the present disclosure provides enhancements for batch mode measurement reporting. A first technique described herein is referred to as “Increasing NR-TimeStamp Priority Reporting.” For this technique, a UE will include the results for the PRS measurements (i.e., the positioning measurements of PRS resources) in the measurement report based on increasing values of the timestamps (e.g., “NR-TimeStamp” IEs **700**) associated with the measurements. Thus, if a timestamp value “T1” (e.g., the SFN value in the “NR-TimeStamp” IE) is greater than a timestamp value “T2,” then the PRS measurement(s) performed at time T1 will be included in the measurement report after the PRS measurement(s) performed at time T2. With this technique, a rollover can be detected when a timestamp in a measurement report has a lower value than the previous/preceding timestamp. For example, if the UE reports, for five PRS measurements, SFN values (i.e., timestamps) of 673, 754, 832, 965, and 43, the location server can determine that the SFN rolled over between the fourth and fifth PRS measurements and act accordingly.

A second technique described herein is referred to as “Decreasing NR-TimeStamp Priority Reporting.” For this technique, a UE will include the results for the PRS measurements in the measurement report based on decreasing values of the timestamps (e.g., “NR-TimeStamp” IEs **700**) associated with the measurements. Thus, if a timestamp value “T1” (e.g., the SFN value in the “NR-TimeStamp” IE) is greater than a timestamp value “T2,” then the PRS measurement(s) performed at time T2 will be included in the measurement report before the PRS measurement(s) performed at time T1. With this technique, a rollover can be detected when a timestamp in a measurement report has a higher value than the previous/preceding timestamp. For example, if the UE reports, for five PRS measurements, SFN values (i.e., timestamps) of 332, 254, 173, 54, 943, and 877, the location server can determine that the SFN rolled over between the third and fourth PRS measurements and act accordingly.

In both the above techniques, the location server will be able to detect the SFN rollover, provided there is only an SFN rollover between subsequent PRS measurements. Accordingly, a third technique described herein provides for using multiple measurement reports. A UE can easily detect whether an SFN rollover will occur, or has occurred, within the measurement window, and can count how many SFN rollovers there will be, or were. Then, for a given measurement request (e.g., an LPP Request Location Information

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message, as at stage **440** of FIG. **4**), the UE can report multiple measurement reports (e.g., LPP Provide Location Information messages, as at stage **460** of FIG. **4**) in the event that the measurement window spans at least one SFN rollover. Specifically, if there are X SFN rollovers inside a measurement window (meaning the measurement window is $X \times 10$ seconds in length), then the UE should transmit at least X separate measurement reports. Reporting the measurement reports should precede or follow the SFN rollover boundary by some threshold amount, specific to UE implementation. In that way, the location server will know to which SFN range the SFNs refer. Note that providing multiple measurement reports for a given measurement request is already defined in LPP, insofar as there may be, for example, more PRS measurements to report than will fit in a single measurement report.

FIG. **9** is a diagram **900** illustrating an example of measurement reporting when a measurement window spans multiple SFN rollovers, according to aspects of the disclosure. In the example of FIG. **9**, time is represented horizontally. A UE receives a measurement request **910** (e.g., an LPP Request Location Information message, as at stage **440** of FIG. **4**) and determines a measurement window **920** based on the information in the measurement request **910**. In the example of FIG. **9**, the measurement window **920** spans three SFN ranges **930** (i.e., SFNs from 0 to 1023) and therefore two SFN rollovers. For each SFN rollover inside the measurement window **920**, the UE transmits a measurement report **940** that includes the PRS measurements taken during the previous SFN range **930**. In addition, the UE transmits a final measurement report **940** at the end of the measurement window **920** to report the PRS measurements taken during the third SFN range **930**.

As a fourth technique described herein, a hyper system frame number (HSFN) field can be added to the “NR-TimeStamp” IE (e.g., “NR-TimeStamp” IE **700**). An HSFN may have a value from 0 to 1023, and each HSFN identifies one of 1024 SFN ranges (e.g., SFN ranges **930**). Thus, adding an HSFN field to the timestamp can uniquely identify the SFN range of the reported SFN, and thereby distinguish the SFN from the SFNs of other SFN ranges. For backwards compatibility with legacy UEs operating according to earlier versions of LPP, this can be an optional field. Whenever there is an SFN rollover inside the measurement window, the UE should add the HSFN field along with SFN and slot number.

FIG. **10** illustrates an example “NR-TimeStamp” IE **1000** that defines the timestamp associated with a UE PRS measurement, according to aspects of the disclosure. As shown in FIG. **10**, an “Nr-HSFN” field has been added to the “NR-TimeStamp” IE that may have a value from 0 to 1023. As noted above, the HSFN field may only be included in the “NR-TimeStamp” IE **1000** if necessary to indicate a rollover of the SFN.

There will be cases where the HSFN is not available or will not be transmitted by the UE’s serving base station. As a first option, the UE can include the HSFN if it is available. As a second option, the UE can include an SFN rollover counter, which will be a positive integer. A value ‘1’ may indicate that an SFN rollover occurred once with respect to the previous measurement report, and a value of ‘2’ may indicate that an SFN rollover occurred twice with respect to the previous measurement report. For example, if the UE only transmits the third measurement report **940** in FIG. **9**, the measurement report would include an SFN rollover counter having a value of ‘2.’ As a third option, the UE can include an absolute timestamp (e.g., a universal time coord-

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inated (UTC) timestamp), if one is available. As a fourth option, the UE can use any other relative or absolute time reporting method.

FIG. **11** illustrates an example method **1100** of wireless positioning, according to aspects of the disclosure. In an aspect, method **1100** may be performed by a UE (e.g., any of the UEs described herein).

At **1110**, the UE performs a plurality of positioning measurements of a corresponding plurality of PRS resources within a measurement window (e.g., measurement window **920**), wherein the measurement window spans at least one SFN rollover. In an aspect, operation **1110** may be performed by the one or more WWAN transceivers **310**, the one or more processors **332**, memory **340**, and/or positioning component **342**, any or all of which may be considered means for performing this operation.

At **1120**, the UE transmits, to a location server (e.g., LMF **270**), at least one measurement report (e.g., measurement report **940**) associated with the measurement window, the at least one measurement report including a plurality of timestamp information elements (e.g., “NR-TimeStamp” IE **700**) corresponding to the plurality of positioning measurements, wherein at least one timestamp information element of the plurality of timestamp information elements indicates the at least one SFN rollover. In an aspect, operation **1120** may be performed by the one or more WWAN transceivers **310**, the one or more processors **332**, memory **340**, and/or positioning component **342**, any or all of which may be considered means for performing this operation.

FIG. **12** illustrates an example method **1200** of wireless positioning, according to aspects of the disclosure. In an aspect, method **1200** may be performed by a UE (e.g., any of the UEs described herein).

At **1210**, the UE performs a plurality of positioning measurements of a corresponding plurality of PRS resources within a measurement window (e.g., measurement window **920**), wherein the measurement window spans at least one SFN rollover. In an aspect, operation **1210** may be performed by the one or more WWAN transceivers **310**, the one or more processors **332**, memory **340**, and/or positioning component **342**, any or all of which may be considered means for performing this operation.

At **1220**, the UE transmits, to a location server (e.g., LMF **270**), one or more measurement reports (e.g., the first and second measurement reports **940** in FIG. **9**) associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range preceding the SFN rollover with which the measurement report is associated. In an aspect, operation **1220** may be performed by the one or more WWAN transceivers **310**, the one or more processors **332**, memory **340**, and/or positioning component **342**, any or all of which may be considered means for performing this operation.

At **1230**, the UE transmits, to the location server, a final measurement report (e.g., the last measurement report **940** in FIG. **9**) associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning measurements performed after a final SFN rollover of the at least one SFN rollover. In an aspect, operation **1230** may be performed by the one or more WWAN transceivers **310**, the one or more processors **332**,

memory 340, and/or positioning component 342, any or all of which may be considered means for performing this operation.

As will be appreciated, a technical advantage of the methods 1100 and 1200 is improved positioning performance by enabling the location server to time-align the UE and the base station(s)/UE(s) transmitting the plurality of PRS resources, as any time misalignment between the measurements will cause performance issues in the positioning calculations.

In the detailed description above it can be seen that different features are grouped together in examples. This manner of disclosure should not be understood as an intention that the example clauses have more features than are explicitly mentioned in each clause. Rather, the various aspects of the disclosure may include fewer than all features of an individual example clause disclosed. Therefore, the following clauses should hereby be deemed to be incorporated in the description, wherein each clause by itself can stand as a separate example. Although each dependent clause can refer in the clauses to a specific combination with one of the other clauses, the aspect(s) of that dependent clause are not limited to the specific combination. It will be appreciated that other example clauses can also include a combination of the dependent clause aspect(s) with the subject matter of any other dependent clause or independent clause or a combination of any feature with other dependent and independent clauses. The various aspects disclosed herein expressly include these combinations, unless it is explicitly expressed or can be readily inferred that a specific combination is not intended (e.g., contradictory aspects, such as defining an element as both an insulator and a conductor). Furthermore, it is also intended that aspects of a clause can be included in any other independent clause, even if the clause is not directly dependent on the independent clause.

Implementation examples are described in the following numbered clauses:

Clause 1. A method of wireless positioning performed by a user equipment (UE), comprising: performing a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmitting, to a location server, at least one measurement report associated with the measurement window, the at least one measurement report including a plurality of timestamp information elements corresponding to the plurality of positioning measurements, wherein at least one timestamp information element of the plurality of timestamp information elements indicates the at least one SFN rollover.

Clause 2. The method of clause 1, wherein the plurality of timestamp information elements each include an SFN value.

Clause 3. The method of clause 2, wherein: the plurality of timestamp information elements is ordered in the at least one measurement report in order of increasing SFN values of the plurality of timestamp information elements, and the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element having a lower SFN value than an SFN value of a preceding timestamp information element of the plurality of timestamp information elements.

Clause 4. The method of clause 2, wherein: the plurality of timestamp information elements is ordered in the at least one measurement report in order of decreasing SFN values of the plurality of timestamp information elements, and the at least one timestamp information element indicating the at

least one SFN rollover comprises the at least one timestamp information element having a higher SFN value than an SFN value of a preceding timestamp information element of the plurality of timestamp information elements.

Clause 5. The method of any of clauses 1 to 4, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including a hyper system frame number (HSFN).

Clause 6. The method of clause 5, wherein only the at least one timestamp information element includes an HSFN.

Clause 7. The method of any of clauses 1 to 6, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including an SFN rollover counter having a value indicating the at least one SFN rollover.

Clause 8. The method of clause 7, wherein: the SFN rollover counter having a first value indicates that the at least one SFN rollover comprises one SFN rollover, and the SFN rollover counter having a second value indicates that the at least one SFN rollover comprises two SFN rollovers.

Clause 9. The method of any of clauses 1 to 8, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including an absolute timestamp.

Clause 10. The method of any of clauses 1 to 9, wherein the plurality of timestamp information elements comprises a plurality of Long-Term Evolution (LTE) positioning protocol (LPP) "NR-TimeStamp" information elements.

Clause 11. A method of wireless positioning performed by a user equipment (UE), comprising: performing a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmitting, to a location server, one or more measurement reports associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range preceding the SFN rollover with which the measurement report is associated; and transmitting, to the location server, a final measurement report associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning measurements performed after a final SFN rollover of the at least one SFN rollover.

Clause 12. The method of clause 11, further comprising: receiving a measurement request from the location server, wherein the one or more measurement reports and the final measurement report are transmitted in response to reception of the measurement request.

Clause 13. The method of clause 12, wherein: the measurement request comprises a Long-Term Evolution (LTE) positioning protocol (LPP) Request Location Information message, and the one or more measurement reports and the final measurement report comprise LPP Provide Location Information messages.

Clause 14. The method of any of clauses 11 to 13, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time before the SFN rollover with which the measurement report is associated.

Clause 15. The method of any of clauses 11 to 14, wherein each measurement report of the one or more measurement

reports is transmitted within a threshold of time after the SFN rollover with which the measurement report is associated.

Clause 16. A user equipment (UE), comprising: a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to: perform a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmit, via the at least one transceiver, to a location server, at least one measurement report associated with the measurement window, the at least one measurement report including a plurality of timestamp information elements corresponding to the plurality of positioning measurements, wherein at least one timestamp information element of the plurality of timestamp information elements indicates the at least one SFN rollover.

Clause 17. The UE of clause 16, wherein the plurality of timestamp information elements each include an SFN value.

Clause 18. The UE of clause 17, wherein: the plurality of timestamp information elements is ordered in the at least one measurement report in order of increasing SFN values of the plurality of timestamp information elements, and the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element having a lower SFN value than an SFN value of a preceding timestamp information element of the plurality of timestamp information elements.

Clause 19. The UE of clause 17, wherein: the plurality of timestamp information elements is ordered in the at least one measurement report in order of decreasing SFN values of the plurality of timestamp information elements, and the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element having a higher SFN value than an SFN value of a preceding timestamp information element of the plurality of timestamp information elements.

Clause 20. The UE of any of clauses 16 to 19, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including a hyper system frame number (HSFN).

Clause 21. The UE of clause 20, wherein only the at least one timestamp information element includes an HSFN.

Clause 22. The UE of any of clauses 16 to 21, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including an SFN rollover counter having a value indicating the at least one SFN rollover.

Clause 23. The UE of clause 22, wherein: the SFN rollover counter having a first value indicates that the at least one SFN rollover comprises one SFN rollover, and the SFN rollover counter having a second value indicates that the at least one SFN rollover comprises two SFN rollovers.

Clause 24. The UE of any of clauses 16 to 23, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including an absolute timestamp.

Clause 25. The UE of any of clauses 16 to 24, wherein the plurality of timestamp information elements comprises a plurality of Long-Term Evolution (LTE) positioning protocol (LPP) "NR-TimeStamp" information elements.

Clause 26. A user equipment (UE), comprising: a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one

transceiver, the at least one processor configured to: perform a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmit, via the at least one transceiver, to a location server, one or more measurement reports associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range preceding the SFN rollover with which the measurement report is associated; and transmit, via the at least one transceiver, to the location server, a final measurement report associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning measurements performed after a final SFN rollover of the at least one SFN rollover.

Clause 27. The UE of clause 26, wherein the at least one processor is further configured to: receive, via the at least one transceiver, a measurement request from the location server, wherein the one or more measurement reports and the final measurement report are transmitted in response to reception of the measurement request.

Clause 28. The UE of clause 27, wherein: the measurement request comprises a Long-Term Evolution (LTE) positioning protocol (LPP) Request Location Information message, and the one or more measurement reports and the final measurement report comprise LPP Provide Location Information messages.

Clause 29. The UE of any of clauses 26 to 28, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time before the SFN rollover with which the measurement report is associated.

Clause 30. The UE of any of clauses 26 to 29, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time after the SFN rollover with which the measurement report is associated.

Clause 31. A user equipment (UE), comprising: means for performing a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and means for transmitting, to a location server, at least one measurement report associated with the measurement window, the at least one measurement report including a plurality of timestamp information elements corresponding to the plurality of positioning measurements, wherein at least one timestamp information element of the plurality of timestamp information elements indicates the at least one SFN rollover.

Clause 32. The UE of clause 31, wherein the plurality of timestamp information elements each include an SFN value.

Clause 33. The UE of clause 32, wherein: the plurality of timestamp information elements is ordered in the at least one measurement report in order of increasing SFN values of the plurality of timestamp information elements, and the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element having a lower SFN value than an SFN value of a preceding timestamp information element of the plurality of timestamp information elements.

Clause 34. The UE of clause 32, wherein: the plurality of timestamp information elements is ordered in the at least one measurement report in order of decreasing SFN values of the plurality of timestamp information elements, and the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element having a higher SFN value than an SFN value of a preceding timestamp information element of the plurality of timestamp information elements.

Clause 35. The UE of any of clauses 31 to 34, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including a hyper system frame number (HSFN).

Clause 36. The UE of clause 35, wherein only the at least one timestamp information element includes an HSFN.

Clause 37. The UE of any of clauses 31 to 36, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including an SFN rollover counter having a value indicating the at least one SFN rollover.

Clause 38. The UE of clause 37, wherein: the SFN rollover counter having a first value indicates that the at least one SFN rollover comprises one SFN rollover, and the SFN rollover counter having a second value indicates that the at least one SFN rollover comprises two SFN rollovers.

Clause 39. The UE of any of clauses 31 to 38, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including an absolute timestamp.

Clause 40. The UE of any of clauses 31 to 39, wherein the plurality of timestamp information elements comprises a plurality of Long-Term Evolution (LTE) positioning protocol (LPP) "NR-TimeStamp" information elements.

Clause 41. A user equipment (UE), comprising: means for performing a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and means for transmitting, to a location server, one or more measurement reports associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range preceding the SFN rollover with which the measurement report is associated; and means for transmitting, to the location server, a final measurement report associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning measurements performed after a final SFN rollover of the at least one SFN rollover.

Clause 42. The UE of clause 41, further comprising: means for receiving a measurement request from the location server, wherein the one or more measurement reports and the final measurement report are transmitted in response to reception of the measurement request.

Clause 43. The UE of clause 42, wherein: the measurement request comprises a Long-Term Evolution (LTE) positioning protocol (LPP) Request Location Information message, and the one or more measurement reports and the final measurement report comprise LPP Provide Location Information messages.

Clause 44. The UE of any of clauses 41 to 43, wherein each measurement report of the one or more measurement

reports is transmitted within a threshold of time before the SFN rollover with which the measurement report is associated.

Clause 45. The UE of any of clauses 41 to 44, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time after the SFN rollover with which the measurement report is associated.

Clause 46. A non-transitory computer-readable medium storing computer-executable instructions that, when executed by a user equipment (UE), cause the UE to: perform a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmit, to a location server, at least one measurement report associated with the measurement window, the at least one measurement report including a plurality of timestamp information elements corresponding to the plurality of positioning measurements, wherein at least one timestamp information element of the plurality of timestamp information elements indicates the at least one SFN rollover.

Clause 47. The non-transitory computer-readable medium of clause 46, wherein the plurality of timestamp information elements each include an SFN value.

Clause 48. The non-transitory computer-readable medium of clause 47, wherein: the plurality of timestamp information elements is ordered in the at least one measurement report in order of increasing SFN values of the plurality of timestamp information elements, and the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element having a lower SFN value than an SFN value of a preceding timestamp information element of the plurality of timestamp information elements.

Clause 49. The non-transitory computer-readable medium of clause 47, wherein: the plurality of timestamp information elements is ordered in the at least one measurement report in order of decreasing SFN values of the plurality of timestamp information elements, and the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element having a higher SFN value than an SFN value of a preceding timestamp information element of the plurality of timestamp information elements.

Clause 50. The non-transitory computer-readable medium of any of clauses 46 to 49, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including a hyper system frame number (HSFN).

Clause 51. The non-transitory computer-readable medium of clause 50, wherein only the at least one timestamp information element includes an HSFN.

Clause 52. The non-transitory computer-readable medium of any of clauses 46 to 51, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including an SFN rollover counter having a value indicating the at least one SFN rollover.

Clause 53. The non-transitory computer-readable medium of clause 52, wherein: the SFN rollover counter having a first value indicates that the at least one SFN rollover comprises one SFN rollover, and the SFN rollover counter having a second value indicates that the at least one SFN rollover comprises two SFN rollovers.

Clause 54. The non-transitory computer-readable medium of any of clauses 46 to 53, wherein the at least one timestamp information element indicating the at least one SFN rollover comprises the at least one timestamp information element including an absolute timestamp.

Clause 55. The non-transitory computer-readable medium of any of clauses 46 to 54, wherein the plurality of timestamp information elements comprises a plurality of Long-Term Evolution (LTE) positioning protocol (LPP) "NR-TimeStamp" information elements.

Clause 56. A non-transitory computer-readable medium storing computer-executable instructions that, when executed by a user equipment (UE), cause the UE to: perform a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and transmit, to a location server, one or more measurement reports associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range preceding the SFN rollover with which the measurement report is associated; and transmit, to the location server, a final measurement report associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning measurements performed after a final SFN rollover of the at least one SFN rollover.

Clause 57. The non-transitory computer-readable medium of clause 56, further comprising computer-executable instructions that, when executed by the UE, cause the UE to: receive a measurement request from the location server, wherein the one or more measurement reports and the final measurement report are transmitted in response to reception of the measurement request.

Clause 58. The non-transitory computer-readable medium of clause 57, wherein: the measurement request comprises a Long-Term Evolution (LTE) positioning protocol (LPP) Request Location Information message, and the one or more measurement reports and the final measurement report comprise LPP Provide Location Information messages.

Clause 59. The non-transitory computer-readable medium of any of clauses 56 to 58, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time before the SFN rollover with which the measurement report is associated.

Clause 60. The non-transitory computer-readable medium of any of clauses 56 to 59, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time after the SFN rollover with which the measurement report is associated.

Those of skill in the art will appreciate that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

Further, those of skill in the art will appreciate that the various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To

clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an ASIC, a field-programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, for example, a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

The methods, sequences and/or algorithms described in connection with the aspects disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in random access memory (RAM), flash memory, read-only memory (ROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), registers, hard disk, a removable disk, a CD-ROM, or any other form of storage medium known in the art. An example storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a user terminal (e.g., UE). In the alternative, the processor and the storage medium may reside as discrete components in a user terminal.

In one or more example aspects, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A storage media may be any available media that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code in the form of instructions or data structures and that can be accessed by a computer. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and

microwave are included in the definition of medium. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. 5
Combinations of the above should also be included within the scope of computer-readable media.

While the foregoing disclosure shows illustrative aspects of the disclosure, it should be noted that various changes and modifications could be made herein without departing from the scope of the disclosure as defined by the appended claims. The functions, steps and/or actions of the method claims in accordance with the aspects of the disclosure described herein need not be performed in any particular order. Furthermore, although elements of the disclosure may be described or claimed in the singular, the plural is contemplated unless limitation to the singular is explicitly stated.

What is claimed is:

1. A method of wireless positioning performed by a user equipment (UE), comprising:

performing a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and

transmitting, to a location server, one or more measurement reports associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range preceding the SFN rollover with which the measurement report is associated; and

transmitting, to the location server, a final measurement report associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning measurements performed after a final SFN rollover of the at least one SFN rollover.

2. The method of claim 1, further comprising:

receiving a measurement request from the location server, wherein the one or more measurement reports and the final measurement report are transmitted in response to reception of the measurement request.

3. The method of claim 2, wherein:

the measurement request comprises a Long-Term Evolution (LTE) positioning protocol (LPP) Request Location Information message, and

the one or more measurement reports and the final measurement report comprise LPP Provide Location Information messages.

4. The method of claim 1, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time before the SFN rollover with which the measurement report is associated.

5. The method of claim 1, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time after the SFN rollover with which the measurement report is associated.

6. A user equipment (UE), comprising:

one or more memories;

one or more transceivers; and

one or more processors communicatively coupled to the one or more memories and the one or more transceiv-

ers, the one or more processors, either alone or in combination, configured to:

perform a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and

transmit, via the one or more transceivers, to a location server, one or more measurement reports associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range preceding the SFN rollover with which the measurement report is associated; and

transmit, via the one or more transceivers, to the location server, a final measurement report associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning measurements performed after a final SFN rollover of the at least one SFN rollover.

7. The UE of claim 6, wherein the one or more processors, either alone or in combination, are further configured to:

receive, via the one or more transceivers, a measurement request from the location server, wherein the one or more measurement reports and the final measurement report are transmitted in response to reception of the measurement request.

8. The UE of claim 7, wherein:

the measurement request comprises a Long-Term Evolution (LTE) positioning protocol (LPP) Request Location Information message, and

the one or more measurement reports and the final measurement report comprise LPP Provide Location Information messages.

9. The UE of claim 6, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time before the SFN rollover with which the measurement report is associated.

10. The UE of claim 6, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time after the SFN rollover with which the measurement report is associated.

11. A user equipment (UE), comprising:

means for performing a plurality of positioning measurements of a corresponding plurality of positioning reference signal (PRS) resources within a measurement window, wherein the measurement window spans at least one system frame number (SFN) rollover; and

means for transmitting, to a location server, one or more measurement reports associated with the measurement window, each measurement report of the one or more measurement reports associated with an SFN rollover of the at least one SFN rollover, each measurement report of the one or more measurement reports including positioning measurements of the plurality of positioning measurements performed during an SFN range preceding the SFN rollover with which the measurement report is associated; and

means for transmitting, to the location server, a final measurement report associated with the measurement window, the final measurement report including positioning measurements of the plurality of positioning

measurements performed after a final SFN rollover of the at least one SFN rollover.

12. The UE of claim **11**, further comprising:

means for receiving a measurement request from the location server, wherein the one or more measurement reports and the final measurement report are transmitted in response to reception of the measurement request.

13. The UE of claim **12**, wherein:

the measurement request comprises a Long-Term Evolution (LTE) positioning protocol (LPP) Request Location Information message, and

the one or more measurement reports and the final measurement report comprise LPP Provide Location Information messages.

14. The UE of claim **11**, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time before the SFN rollover with which the measurement report is associated.

15. The UE of claim **11**, wherein each measurement report of the one or more measurement reports is transmitted within a threshold of time after the SFN rollover with which the measurement report is associated.

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